



# Dynamics of singularly perturbed sliding flow in Filippov systems

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Received: 19 July 2025 / Revised: 29 October 2025 / Accepted: 11 November 2025 / Published online: 22 December 2025  
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## Abstract

In this article, we present an analysis of the effects of singular perturbations on the sliding motion in Filippov systems. We show that singular perturbations may lead to qualitatively distinct topologies of phase space on the switching manifold, which we classify into six distinct topologies. Five of these topologies imply that singularly perturbed trajectory includes a segment (or segments) of sliding, and one topology, which we study here, implies the evolution characterised by switchings between trajectory segments along the switching surface, but without any sliding. In particular, we show that in the case of  $n$ -dimensional Filippov systems with one switching surface and  $m$ -dimensional fast dynamics, which plays the role of a stable singular perturbation, the flow follows sliding motion of the reduced system, but the perturbation becomes time dependent and may create a micro chaotic behaviour. However, a significant change in the flow direction is not possible. In the particular case of 1-dimensional fast dynamics, the singular perturbation implies regular perturbation of  $\mathcal{O}(\varepsilon)$  of the perturbed flow at the points of switching between vector fields.

**Keywords** Filippov systems · Sliding flow · Singular perturbations · Slow-fast dynamics

**Mathematics Subject Classification** MSC 34D15 · MSC 37D99 · MSC 37N35

## 1 Introduction

In the process of mathematical modelling, the standard procedure of reducing a given problem's complexity may result in a mathematical framework with discontinuous nonlinearities. To give just a few examples, systems characterised by impacts, switchings, and/or by events which on the macroscopic scale occur almost instantaneously [1, 2, 15, 16, 19, 20] are all examples where the framework of discontinuous systems is of particular relevance. To such a class of systems belong so-called *Filippov systems* [6, 21]. That is dynamical systems with phase space partitioned into non-empty adjacent regions where in each region a distinct set of smooth ordinary differential equations (ODEs) governs the dynamics. In the simplest case we have just two regions, say  $\mathcal{H}_+$

and  $\mathcal{H}_-$  with a boundary  $\mathcal{H}$  between these regions, called a *switching manifold* (or *surface*). We may then consider Filippov systems, differential equations of the form

$$\dot{x} = \begin{cases} f_+(x) & \text{if } h(x) > 0, \\ f_-(x) & \text{if } h(x) < 0, \end{cases} \quad (1)$$

with a boundary  $\mathcal{H} = \{h(x) = 0\}$ , where  $f_{\pm}$  are vector fields obtained from ODEs describing the dynamics in  $\mathcal{H}_{\pm}$  and  $h(x)$  is a scalar valued multivariable smooth function. A particular feature of Filippov systems is the possibility that in some non-empty region  $\mathcal{H}_{ss}$  of the boundary  $\mathcal{H}$  both vector fields point towards  $\mathcal{H}$ . In this case the directional derivatives satisfy

$$\begin{aligned} \frac{dh(X_+^t(x))}{dt} &= h_x f_+ < 0, \\ \frac{dh(X_-^t(x))}{dt} &= h_x f_- > 0 \text{ for } x \in \mathcal{H}_{ss} \subseteq \mathcal{H}, \end{aligned}$$

where  $X_{\pm}^t$  are flows generated by the vector fields  $f_{\pm}$  (all functions are taken at their argument  $x$ , and the subscript denotes the partial derivative). Hence, for points  $x \in \mathcal{H}_{ss}$  one can not define a flow of (1) by either following  $X_+^t$  or

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$X_-^t$ . A convention introduced by [6, 21] is to consider a *sliding vector field*, which generates a *sliding flow*  $X_s^t$  on  $\mathcal{H}_{ss}$ . The sliding vector field is a convex combination of  $f_{\pm}$  of the form

$$\dot{x} = f_s(x) = f_+ + \alpha(f_- - f_+), \text{ where} \quad (2)$$

$$\alpha = -\frac{h_x f_+}{h_x(f_- - f_+)}, \alpha \in [0, 1].$$

As we have already indicated in [18], an important question to consider is how the sliding vector field is affected by perturbations. Regular (that is,  $C^1$ -small) perturbations applied to  $f_{\pm}$  and  $h$  do not qualitatively alter the system's dynamics provided some generic hyperbolicity and transversality conditions hold, similar to the conditions and theorems on the persistence and stability of invariant sets in smooth dynamical systems [5]. An open question is how additional degrees of freedom that are stable and rapidly decaying affect the sliding flow. Simple examples of such perturbations in modelling of systems with discontinuities are when one replaces a contact to a rigid wall by a finite stiffness, mass and damping in mechanical models, or takes into account a previously neglected small capacitance of an electrical element in a circuit with a switch. Mathematically these additional degrees of freedom are called *stable singular perturbations*. The reverse approach is typically used to reduce the system dimension of an ODE model. One replaces rapidly converging subsystems of an ODE model by their equilibrium value, which changes slowly (*quasistatically*), depending on the slower subsystems of the ODE.

It is easy to demonstrate that such stable singular perturbations can change a sliding vector field into a rapid sequence of switching back-and-forth between vector fields (see [18] for minimal examples). Thus, no direct equivalent of the persistence results for smooth vector fields by [5] can possibly hold in general. Fridman [7, 8] showed that, if the switching function depends only on the slow variables, periodic orbits persist under singular perturbations, but they acquire a small boundary layer. If one allows that the switching decision function depends on fast variables, this may lead to a period-adding cascade or small-scale chaos that is not present in the reduced system [18].

The aim of this paper is to close an existing gap in the theory of singularly perturbed Filippov systems and provide an answer to the question what is the effect of stable singular perturbation on sliding motion in Filippov systems. Our exploration how the Filippov convention interacts with additional stable degrees of freedom is complementary to recent work by Kristiansen and Hogan [11–13]. Their work starts from the assumption that the discontinuity in the right-hand side of (1) is the singular limit (“ $\varepsilon = 0$ ”) of an underlying smooth singularly perturbed system (“ $\varepsilon > 0$ ”). Then they study the dynamics for small  $\varepsilon > 0$  using blow-up tech-

niques [14] to observe limits of the dynamics and see if these limits are consistent with the Filippov convention.

In the current work, we exclude the possible effects of the presence of so-called  $U$ -singularity (also termed as two-fold singularity) in Filippov systems of interests, which, among other works, have been studied in [3].

The outline of the rest of the paper is as follows. Section 2 introduces the type of singular perturbations to Filippov systems we study, contrasting the *full system* ( $\varepsilon > 0$ ) and the *reduced system* ( $\varepsilon = 0$ ). Section 3 performs a rescaling, zooming into the neighborhood of a point near the switching surface of the reduced system, a coordinate transformation and a truncation to leading order in  $\varepsilon$ , after which  $\varepsilon$  disappears as parameter. The main results are then presented in Sec. 4, and in Sec. 5 and 6 where they are encapsulated in Lemma 4, Theorem 6 and Theorem 7. Section 7 illustrates our findings with a numerical example. Finally Section 8 concludes the paper.

## 2 Filippov systems and fast subsystems — main results

*Slow-fast vector fields with discontinuity and Filippov convention* We consider ODEs for  $x \in \mathbb{R}^n$  with a discontinuous right-hand side, which interact with a *fast subsystem* (an ODE for  $y \in \mathbb{R}^m$  on a faster time scale):

$$\text{slow subsystem: } \dot{x} = \begin{cases} f_+(x, y, \varepsilon) & \text{if } h(x, y, \varepsilon) > 0, \\ f_-(x, y, \varepsilon) & \text{if } h(x, y, \varepsilon) < 0, \end{cases} \quad (3)$$

$$\text{fast subsystem: } \varepsilon \dot{y} = g(x, y, \varepsilon), \quad (4)$$

where  $\varepsilon$  is small, right-hand sides  $f_{\pm} : \mathbb{R}^{n+m+1} \mapsto \mathbb{R}^n$ ,  $g : \mathbb{R}^{n+m+1} \mapsto \mathbb{R}^m$ , and *switching function*  $h : \mathbb{R}^{n+m+1} \mapsto \mathbb{R}$  are sufficiently smooth in some region in  $\mathbb{R}^{n+m+1}$  and the dot symbol denotes differentiation with respect to  $t$ . We permit that on the zero level set of  $h$  (the *switching manifold*, a codimension-1 hypersurface),

$$\mathcal{H}_{\varepsilon} = \{(x, y) \in \mathbb{R}^{n+m} : h(x, y, \varepsilon) = 0\},$$

$f_+(x, y, \varepsilon)$  and  $f_-(x, y, \varepsilon)$  may differ, such that the right-hand side in (3) is discontinuous in  $\mathcal{H}_{\varepsilon}$ . The variable  $y$  is called the *fast variable*, while  $x$  is the *slow variable*.

We assume that for fixed slow variable  $x$  there exists a stable equilibrium  $y_c(x)$  ( $y_c : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ) of the fast subsystem after rescaling time by  $\varepsilon$ ,  $\dot{y} = g(x, y, \varepsilon)$  at  $\varepsilon = 0$  for each fixed  $x$ . The map  $y_c$  is defined implicitly by the equation

$$g(x, y_c(x), 0) = 0. \quad (5)$$

The set of equilibria of the fast subsystem in rescaled time is called the *critical manifold*

$$\begin{aligned}\mathcal{M}_0 &= \{(x, y) \in \mathbb{R}^{n+m} : g(x, y, 0) = 0\} \\ &= \{(x, y) \in \mathbb{R}^{n+m} : y = y_c(x)\}.\end{aligned}$$

This critical manifold  $\mathcal{M}_0$  is a graph over the slow variable. The following assumption makes the request for stability of  $y_c$  precise.

**Assumption 1** (Transversal stability) We assume that the equilibria in  $\mathcal{M}_0$  are uniformly exponentially stable for fixed  $x$  (using  $\text{spec } A$  for the spectrum of a matrix  $A$ ):  $\text{Re spec } g_y(x, y_c(x), 0) < -c_{\text{stab}} < 0$  for all  $x$ .

*Brief review of smooth slow-fast singular perturbation theory* Let us briefly for reference summarize the classical result by Fenichel [5] for the scenario of smooth slow-fast systems. For (3), (4) this is the case where  $f_+ \equiv f_- =: f$  in (3) such that  $h$  is irrelevant. In the smooth case Assumption 1 permits one to reduce the study of the long-time dynamics of the full slow-fast system (3), (4) near the critical manifold  $\mathcal{M}_0$  to a study of dynamics of the *reduced system*,  $\dot{x} = f(x, y_c(x), 0)$ .

More precisely, classical singular perturbation theory by Fenichel [5] ensures that the full system has an invariant transversally stable manifold  $\mathcal{M}_{\text{slow}} = \{(x, y) : y = y_{\text{sl}}(x, \varepsilon)\}$ . On the manifold  $\mathcal{M}_{\text{slow}}$  the dynamics of  $x$  is governed by the exact reduced ODE

$$\dot{x} = f(x, y_{\text{sl}}(x, \varepsilon), \varepsilon) \quad (6)$$

(with  $X_{\text{re}}^t(x, \varepsilon)$  a time- $t$  map starting from  $x$ ). Furthermore, every initial condition  $(x_0, y_0)$  near  $\mathcal{M}_{\text{slow}}$  there exists a  $x_{\text{fib}}(x_0, y_0, \varepsilon)$  such that the full solution  $(x(t), y(t))$  satisfies

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} - \begin{bmatrix} X_{\text{re}}^t(x_{\text{fib}}(x_0, y_0, \varepsilon), \varepsilon) \\ y_{\text{sl}}(X_{\text{re}}^t(x_{\text{fib}}(x_0, y_0, \varepsilon), \varepsilon), \varepsilon) \end{bmatrix} \sim \exp(-c_{\text{stab}}t). \quad (7)$$

Consequently, the long-time behavior of all trajectories near  $\mathcal{M}_{\text{slow}}$ , and, hence, near the critical manifold  $\mathcal{M}_0$  is completely described up to exponentially decaying terms (so-called *boundary layers*) by the exact reduced ODE (6) and its flow  $X_{\text{re}}^t$ .

The slow manifold  $\mathcal{M}_{\text{slow}}$  is in general not unique. However, for smooth  $f$  and  $g$  its graph  $y_{\text{sl}}(x, \varepsilon)$  has a unique expansion in  $\varepsilon$ . Its zero order term equals  $y_c(\varepsilon)$ . Its first-order expansion is

$$y_{\text{sl}}(x, \varepsilon) = y_c(x) - \varepsilon[g_y]^{-1}[g_\varepsilon + [g_y]^{-1}g_x f] + O(\varepsilon^2), \quad (8)$$

where  $g_x$ ,  $g_y$  and  $g_\varepsilon$  are the respective partial derivatives of  $g$  and all terms on the right-hand side are evaluated in  $(x, y_c(x), 0)$ .

*Assumptions for the singularly perturbed Filippov system* Inserting the graph  $y_c$  of the critical manifold as the zero-order expansion of  $y_{\text{sl}}$  into the Filippov system (3), (4), we obtain a *reduced system* on the critical manifold  $\mathcal{M}_0$ , which is also of Filippov type in  $\mathbb{R}^n$ ,

$$\dot{x} = \begin{cases} f_+(x, y_c(x), 0) & \text{if } h_{\text{rd}}(x) > 0, \\ f_-(x, y_c(x), 0) & \text{if } h_{\text{rd}}(x) < 0, \end{cases} \quad \text{where } h_{\text{rd}}(x) = h(x, y_c(x), 0). \quad (9)$$

In our envisioned scenario a modeller will consider this reduced system (9), because they have neglected stable fast dynamics, replacing the fast ODE (4),  $\varepsilon \dot{y} = g(x, y, \varepsilon)$ , by the zero-order quasi-steady-state approximation  $y = y_c(x)$ , and set  $\varepsilon = 0$ . If a trajectory  $x(t)$  of (9) stays in one of the subdomains  $\{x : \pm h_{\text{rd}}(x) > 0\}$  for a time of order  $O(1)$  then we can apply the smooth singular perturbation theory to deduce that for sufficiently small  $\varepsilon$  the full flow from any initial condition  $(x_0, y_0) \approx (x(0), y_c(x(0)))$  can be approximated by  $X_{\text{re}}^t$  according to (7), where we use  $f_\pm$  in the expansion of  $y_{\text{sl}}$  for  $f$ . The exact reduced flow  $X_{\text{re}}^t$  in turn is a  $O(\varepsilon)$  perturbation of the flow (9) in its subdomain.

It is less clear what happens to trajectories that approach the discontinuity surface  $\{x : h_{\text{rd}}(x) = 0\}$ . So, we consider the dynamics of  $(x, y)$  near a point  $(x_0, y_c(x_0)) \in \mathcal{H}_{\text{rd}}$  on the intersection of switching and critical manifolds. This means that we assume that  $x_0$  is a point on the discontinuity surface of the reduced system (9):

$$\begin{aligned}(x_0, y_c(x_0)) &\in \mathcal{H}_0 \cap \mathcal{M}_0 \text{ s.t.} \\ h_{\text{rd}}(x_0) &= h(x_0, y_c(x_0), 0) = 0, \text{ or,} \\ \text{equivalently, } h(x_0, y_0, 0) &= 0 \text{ and } g(x_0, y_0, 0) = 0.\end{aligned} \quad (10)$$

We assume that  $x_0$  is in an attracting sliding region of the reduced system (9).

**Assumption 2** (Attracting sliding in reduced system) We assume that

$$h_{\text{rd},x} f_+ < 0 < h_{\text{rd},x} f_-, \text{ where } h_{\text{rd},x} = h_x - h_y[g_y]^{-1}g_x. \quad (11)$$

The coefficients in (11) are evaluated at  $(x_0, y_c(x_0), 0)$ . The reduced system (9) thus has an attracting sliding vector field in  $\{x : h_{\text{rd}}(x) = 0\}$  near  $x_0$ :

$$\begin{aligned}\dot{x} &= f_{\text{rd},s}(x) := f_+ + \alpha_{\text{rd}}(f_- - f_+), \text{ where} \\ \alpha_{\text{rd}} &= -\frac{h_{\text{rd},x} f_+}{h_{\text{rd},x}(f_- - f_+)},\end{aligned} \quad (12)$$

evaluating all coefficients in  $x$  and  $(x, y_c(x), 0)$ . Assumption 2 implies that the vector  $h_x - h_y[g_y]^{-1}g_x \neq 0$  in  $(x_0, y_c(x_0), 0)$ . Hence, the intersection of the switching manifold  $\mathcal{H}_0$  and the critical manifold  $\mathcal{M}_0$  is transversal in  $(x_0, y_c(x_0))$  at  $\varepsilon = 0$ , such that

$$\begin{aligned}\mathcal{H}_{\text{rd}} &= \mathcal{H}_0 \cap \mathcal{M}_0 \\ &= \{(x, y) \in \mathbb{R}^{n+m} : h(x, y, 0) = 0, g(x, y, 0) = 0\}\end{aligned}$$

is locally a smooth manifold of dimension  $n - 1$  near  $(x_0, y_c(x_0))$ .

*The  $O(\varepsilon)$  gap between the slow manifolds* We observe that the exact slow manifolds  $y_{\text{sl}, \pm}(x, \varepsilon)$  that are attracting nearby dynamics for the respective regions are different. Their difference is generically of order  $\varepsilon$  in  $x_0$ :

$$y_{\text{sl}, +}(x_0, \varepsilon) - y_{\text{sl}, -}(x_0, \varepsilon) = -\varepsilon[g_y^0]^{-2}g_x^0[f_+^0 - f_-^0] + O(\varepsilon^2)$$

(all coefficients evaluated at  $(x_0, y_c(x_0), 0)$ ). So, we expect, for example, that a trajectory  $(x(t), y(t))$  of the full system (3), (4) that has spent time of order 1 in  $\{h(x, y, \varepsilon) > 0\}$  will have  $y(t_0)$  close  $y_{\text{sl}, +}(x_0, \varepsilon)$  when  $x(t_0)$  equals  $x_0$  (so, when approaching the discontinuity). It will then follow a different vector field after reaching  $\{h = 0\}$ . Thus, we expect that  $(x(t), y(t))$  will follow boundary layers of size  $O(\varepsilon)$  for  $t > t_0$ .

*Main results* Which transitions between  $f_-$ ,  $f_+$  and sliding trajectories starting from  $(x, y)$  are possible on time scale larger than  $O(\varepsilon)$  depends on the ordering of the four scalar quantities

$$T_{\pm} := h_x^0 f_{\pm}^0, \quad S_{\pm} := h_y^0 [g_y^0]^{-1} g_x^0 f_{\pm}^0 \quad (13)$$

(all coefficients evaluated at  $(x_0, y_c(x_0), 0)$ ). When full system (3), (4) is restricted to half space  $\{\pm h(x, y, \varepsilon) > 0\}$  it will approach the invariant manifold  $\mathcal{M}_{\text{slow}, \pm} = \{(x, y) : x = y_{\text{sl}, \pm}(x, \varepsilon)\}$ , which deviates from  $y_c(x)$  by order  $\varepsilon$ . The quantities  $S_{\pm}$  measure in which direction along  $\mathcal{H}_0$  the intersection between  $\mathcal{H}_0$  and  $\mathcal{M}_0$  is split by this deviation. The quantities  $T_{\pm}$  show where the full vector fields  $(f_{\pm}, g/\varepsilon)$  are tangent to the switching manifold  $\mathcal{H}_0$ , which can occur near  $(x_0, y_c(x_0))$  despite Assumption 2 (note the presence of  $h_{\text{rd}, x}(x_0)$  in (11), but of  $h_x^0$  in (13)). Assumption 2 is equivalent to  $T_+ < S_+$  and  $S_- < T_-$ . The six possible orderings of  $S_{\pm}$  and  $T_{\pm}$  compatible with Assumption 2 are listed in Table 1 and illustrated in Fig. 1. Of these two are symmetrically related, two permit only attracting sliding near  $(x_0, y_c(x_0))$ , and one (the case  $S_- < T_- < T_+ < S_+$ ) permits only repelling sliding. For the case  $S_- < T_- < T_+ < S_+$  with repelling sliding we establish the following perturbation result.

**Theorem 3** (Smallness of stable fast perturbations) *Assume that the slow-fast system (3), (4) satisfies Assumptions 1 and*

*2 near  $(x_0, y_c(x_0))$  and assume that  $S_- < T_- < T_+ < S_+$ . There exists a perturbation bound  $\rho_f > 0$  such that for all bounds  $\rho_x, \rho_y, \rho_t > 0$  we can find an  $\varepsilon_{\text{max}} > 0$  such that the following holds.*

*If  $|\delta_x| \leq \rho_x$ ,  $|\delta_y| \leq \rho_y$ ,  $t \in [0, \rho_t]$  and  $\varepsilon \in (0, \varepsilon_{\text{max}})$  then the trajectory  $(x(t), y(t))$  of (3), (4) starting from  $(x_0 + \varepsilon\delta_x, y_c(x_0) + \varepsilon\delta_y)$  without repelling sliding satisfies*

$$\begin{aligned}& \begin{bmatrix} x(\varepsilon t) \\ y(\varepsilon t) \end{bmatrix} - \begin{bmatrix} x(0) \\ y(0) \end{bmatrix} \\ &= \varepsilon t \begin{bmatrix} f_{\text{rd}, s}(x_0) \\ [g_y^0]^{-1} g_x^0 f_{\text{rd}, s}(x_0) \end{bmatrix} + \varepsilon r(t, \delta_x, \delta_y, \varepsilon) \quad (14)\end{aligned}$$

*for some continuous function  $r$  bounded by  $|r(t, x, y, \varepsilon)| \leq \rho_f$  for all  $t \in \mathbb{R}$ ,  $\varepsilon \in [0, \varepsilon_{\text{max}}]$ ,  $|x| \leq \rho_x$ ,  $|y| \leq \rho_y$ .*

The theorem states that the reduced sliding flow (12) with right-hand side  $f_{\text{rd}, s}$  is a good approximation on time scales greater than  $\varepsilon$ . This choice is possible in the theorem, since the bound  $\rho_f$  of the perturbation scaled by  $\varepsilon$  is independent of the time scale  $\rho_t$  scaled by  $\varepsilon$ . Thus, we can choose  $t \gg 1$  such that still  $\varepsilon t \ll 1$ . In that case the drift of order  $\varepsilon t$  dominates the perturbation  $\varepsilon r$ . Once  $\varepsilon t$  is no longer small, the drift term, which is evaluated at  $(x_0, y_c(x_0))$  is no longer approximately constant. The perturbation depends on  $t$  and is only continuous so can be rapidly fluctuating. Informally, one could phrase the result of Theorem 3 as  $x(t)$  for the full system (3), (4) satisfying

$$\dot{x}(t) = f_{\text{rd}, s}(x(t)) + \tilde{\varepsilon} \tilde{r}(t/\varepsilon) + O(\varepsilon), \quad (15)$$

where  $\varepsilon \ll \tilde{\varepsilon} \ll 1$  and  $\tilde{r}$  is bounded but fluctuates rapidly (and will depend on quantities not shown in (15), such as  $y(0)$ ). One obtains (15) by dividing (14) by  $\varepsilon t$ , assuming  $t$  large and  $\varepsilon$  small, such that  $\tilde{\varepsilon} = 1/t$ . In Sect. 4 the perturbation is constructed as the trajectory of an approximately piecewise affine map for the scaled-up fast variable  $\delta_y$ . This map can be chaotic, such that we may have small “micro chaos” around the sliding flow, similar to what is observed when one digitizes control inputs [4, 9, 10]. Small-scale fluctuations of the type in (14) were also constructed as a generic feature of (3), (4) when it is near a grazing bifurcation [18].

Our second main result, fully stated in Theorem 7 claims that for one-dimensional fast dynamics ( $n_y = 1$ ) this micro chaos cannot occur. For  $n_y = 1$ , the small-scale dynamics of  $\delta_y$  is governed by a monotone one-dimensional second return map to the switching surface in the repelling sliding case (always intersecting the switching surface transversally). Thus, its dynamics will converge to a stable small-scale periodic motion on time scales  $1 \ll t \ll 1/\varepsilon$ . Consequently, this case reduces to periodic averaging with a periodic motion on the fast time scale, such that the perturbation  $\tilde{r}$  in (15) will be a regular perturbation of the form  $\tilde{r}(x(t), \varepsilon)$ . Moreover,

the one-dimensional map approaches the identity map as the quantity  $T_+ - T_- = h_x^0(f_+^0 - f_-^0)$  (which depends on  $x_0$ ) approaches 0.

### 3 Rescaling and reduction to leading order in small parameter

We will study the dynamics near a point  $(x_0, y_c(x_0)) \in \mathcal{H}_{rd}$ , an arbitrary point on the switching manifold of the reduced system, where we have attracting sliding for the reduced system (so, (11) is satisfied).

#### 3.1 Zoom-in in space and slow-down of time

We now rescale the state variables

$$x_{old} = x_0 + \varepsilon x_{new}, \quad y_{old} = y_c(x_0) + \varepsilon y_{new}, \quad t_{old} = \varepsilon t_{new}. \quad (16)$$

Substituting the zoom-in (16) into system equations (3), (4), we obtain (removing subscripts “new” and “old”)

$$\dot{x} = \begin{cases} f_+(x_0 + \varepsilon x, y_c(x_0) + \varepsilon y, \varepsilon) & \text{if } h(x_0 + \varepsilon x, y_c(x_0) + \varepsilon y, \varepsilon) > 0, \\ f_-(x_0 + \varepsilon x, y_c(x_0) + \varepsilon y, \varepsilon) & \text{if } h(x_0 + \varepsilon x, y_c(x_0) + \varepsilon y, \varepsilon) < 0, \end{cases} \quad (17)$$

$$\varepsilon \dot{y} = g(x_0 + \varepsilon x, y_c(x_0) + \varepsilon y, \varepsilon). \quad (18)$$

Note that using the above rescaling means that a distance of  $\mathcal{O}(\varepsilon)$  in  $(x_{old}, y_{old})$  from  $(x_0, y_c(x_0))$  corresponds to a distance of  $\mathcal{O}(1)$  in  $(x_{new}, y_{new})$  from  $(0, 0)$ . In the new coordinates  $(x_{new}, y_{new}) = (0, 0)$  corresponds to  $(x_{old}, y_{old}) = (x_0, y_c(x_0))$ .

#### 3.2 Truncation to leading order in $\varepsilon$ and shift of origin

Expanding the full system (17)–(18) to leading order in  $\varepsilon$  we get

$$\dot{x} = \begin{cases} f_+^0 + \mathcal{O}(\varepsilon) & \text{if } h_x^0 x + h_y^0 y + h_\varepsilon^0 + \mathcal{O}(\varepsilon) > 0, \\ f_-^0 + \mathcal{O}(\varepsilon) & \text{if } h_x^0 x + h_y^0 y + h_\varepsilon^0 + \mathcal{O}(\varepsilon) < 0, \end{cases} \quad (19)$$

$$\dot{y} = g_x^0 x + g_y^0 y + g_\varepsilon^0 + \mathcal{O}(\varepsilon), \quad \text{where} \quad f_\pm^0 = f_\pm(x_0, y_c(x_0), 0), \quad (20)$$

and  $h_x^0 \in \mathbb{R}^{1 \times n}$ ,  $h_y^0 \in \mathbb{R}^{1 \times m}$ ,  $h_\varepsilon^0 \in \mathbb{R}^{1 \times 1}$ ,  $g_x^0 \in \mathbb{R}^{m \times n}$ ,  $g_y^0 \in \mathbb{R}^{m \times m}$  and  $g_\varepsilon^0 \in \mathbb{R}^{m \times 1}$  are the Jacobian matrices evaluated at  $(x_0, y_c(x_0), 0)$ . Correspondingly, the quantity

$$h_{rd,x}^0 = h_x^0 - h_y^0 [g_y^0]^{-1} g_x^0, \quad (21)$$

is the normal to the switching surface in  $(x_0, y_c(x_0))$  of the reduced system (9), which is non-zero as we assume that the

expansion point is on the switching manifold of the reduced system by assumption (11). By shifting the rescaled  $x$  and  $y$  coordinates we may also eliminate the constant terms in the right-hand side expansion ( $g_\varepsilon^0$ ) and the switching function ( $h_\varepsilon^0$ ):

$$x_{old} = x_{new} + \delta_x, \quad y_{old} = y_{new} + \delta_y.$$

The shifts  $\delta_x$  and  $\delta_y$  have to satisfy the affine system of  $m+1$  equations

$$0 = h_x^0 \delta_x + h_y^0 \delta_y + h_\varepsilon^0, \quad 0 = g_x^0 \delta_x + g_y^0 \delta_y + g_\varepsilon^0,$$

for which one possible solution is

$$\delta_x = \delta_h h_{rd,x}^T, \quad \delta_y = -[g_y^0]^{-1} [g_x^0 \delta_h h_{rd,x}^T + g_\varepsilon^0],$$

$$\text{with } \delta_h = \frac{h_y^0 [g_y^0]^{-1} g_\varepsilon^0 - h_\varepsilon^0}{h_{rd,x} h_{rd,x}^T},$$

(this choice of  $\delta_x$  shifts orthogonal to the switching surface). In the new (zoomed-in, slowed-down, and shifted) coordinates, the full system of equation has the simplified form

$$\dot{x} = \begin{cases} f_+^0 + \mathcal{O}(\varepsilon) & \text{if } h_x^0 x + h_y^0 y + \mathcal{O}(\varepsilon) > 0, \\ f_-^0 + \mathcal{O}(\varepsilon) & \text{if } h_x^0 x + h_y^0 y + \mathcal{O}(\varepsilon) < 0, \end{cases} \quad (22)$$

$$\dot{y} = g_x^0 x + g_y^0 y + \mathcal{O}(\varepsilon), \quad (23)$$

where the two  $\mathcal{O}(\varepsilon)$  terms in the switching condition are identical. We observe that this shift removes the derivatives  $g_\varepsilon^0$  and  $h_\varepsilon^0$  with respect to  $\varepsilon$ .

**Truncated reduced system** The truncated reduced system is the piecewise constant flow

$$\dot{x} = \begin{cases} f_+^0 & \text{if } h_{rd,x}^0 x > 0, \\ f_-^0 & \text{if } h_{rd,x}^0 x < 0, \end{cases} \quad (24)$$

$$0 = g_x^0 x + g_y^0 y \quad (\text{thus, } y = -[g_y^0]^{-1} g_x^0 x). \quad (25)$$

In particular, we observe that the condition  $h_{rd,x}^0 f_+^0 < 0 < h_{rd,x}^0 f_-^0$  for attracting sliding for the truncated reduced flow (24), (25), spelled out

$$h_{rd,x}^0 f_+^0 = [h_x^0 - h_y^0 [g_y^0]^{-1} g_x^0] f_+^0 < 0$$

$$< [h_x^0 - h_y^0 [g_y^0]^{-1} g_x^0] f_-^0 = h_{rd,x}^0 f_-^0, \quad (26)$$

also implies sliding of the reduced flow of (22), (23) for all small  $\varepsilon$  and all  $x, y$  of order 1. As we assume that the sliding condition (26) holds, (24) has a constant sliding vector field on the set

$$\mathcal{H}_{\text{rd},x} := \{h_{\text{rd},x}^0 = 0\},$$

(the  $x$ -components of the discontinuity set  $\mathcal{H}_{\text{rd}}$  in the new coordinates) of the form

$$\begin{aligned} \dot{x} &= f_{\text{rd},s}^0 = (1 - \alpha_{\text{rd}}^0) f_+^0 + \\ \alpha_{\text{rd}}^0 f_-^0 &= f_+^0 + \alpha_{\text{rd}}^0 (f_-^0 - f_+^0), \end{aligned} \quad (27)$$

where  $\alpha_{\text{rd}}^0 \in [0, 1]$  is determined by the condition that  $f_{\text{rd},s}^0$  is in  $\mathcal{H}_{\text{rd},x}$ , satisfying  $h_{\text{rd},x}^0 f_{\text{rd},s}^0 = 0$ :

$$\alpha_{\text{rd}}^0 = \frac{h_{\text{rd},x}^0 f_+^0}{h_{\text{rd},x}^0 (f_+^0 - f_-^0)}. \quad (28)$$

So the constant sliding vector field  $f_{\text{rd},s}^0$  for the truncated reduced system (24) equals

$$\begin{aligned} f_{\text{rd},s}^0 &= f_+^0 - \frac{h_{\text{rd},x}^0 f_+^0}{h_{\text{rd},x}^0 (f_-^0 - f_+^0)} (f_-^0 - f_+^0) \\ &= \frac{f_+^0 h_{\text{rd},x}^0 f_-^0 - f_-^0 h_{\text{rd},x}^0 f_+^0}{h_{\text{rd},x}^0 (f_-^0 - f_+^0)} \\ &= \frac{f_+^0 (h_x^0 f_-^0 - h_y^0 [g_y^0]^{-1} g_x^0 f_-^0) - f_-^0 (h_x^0 f_+^0 - h_y^0 [g_y^0]^{-1} g_x^0 f_+^0)}{h_x^0 (f_-^0 - f_+^0) - h_y^0 [g_y^0]^{-1} g_x^0 (f_-^0 - f_+^0)}. \end{aligned} \quad (29)$$

The reduced system after truncation, (27), has, thus, a parallel flow in direction  $f_{\text{rd},s}^0$ .

*Truncated full system* Neglecting terms of order  $\varepsilon$  the truncated full system is

$$\dot{x} = \begin{cases} f_+^0 & \text{if } h_x^0 x + h_y^0 y > 0, \\ f_-^0 & \text{if } h_x^0 x + h_y^0 y < 0, \end{cases} \quad \dot{y} = g_x^0 x + g_y^0 y. \quad (31)$$

We change the  $y$  coordinate such that it moves with  $x$ , eliminating the  $x$  dependence in the  $\dot{y}$  part of (31):  $y_{\text{new}} = -g_y^0 y_{\text{old}} - g_x^0 x$ . In new coordinates (dropping subscript “new”) the full truncated system satisfies

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{cases} F_+ = \begin{bmatrix} f_+^0 \\ g_y^0 y - g_x^0 f_+^0 \end{bmatrix} & \text{if } h_{\text{rd},x}^0 x - h_y^0 [g_y^0]^{-1} y > 0, \\ F_- = \begin{bmatrix} f_-^0 \\ g_y^0 y - g_x^0 f_-^0 \end{bmatrix} & \text{if } h_{\text{rd},x}^0 x - h_y^0 [g_y^0]^{-1} y < 0, \end{cases} \quad (32)$$

and the discontinuity manifold in these new coordinates is the subspace

$$\partial \mathcal{H}_0 = \left\{ (x, y) : h_{\text{rd},x}^0 x = h_y^0 [g_y^0]^{-1} y \right\}$$

(the linearization of  $\mathcal{H}_0$  in  $(x_0, y_c(x_0), 0)$  with respect to  $x$  and  $y$  in  $\varepsilon = 0$ ). As long as the full truncated system (32)

stays in the respective domain for  $F_{\pm}$ , the affine subspace (note that the new coordinate has a prefactor  $-g_y^0$  compared to the original  $y$ )

$\mathcal{M}_{\text{slow},\pm} = \{(x, y) : y = y_{\text{sl},\pm}, x \text{ arbitrary}\}$ , where

$$y_{\text{sl},\pm} := [g_y^0]^{-1} g_x^0 f_{\pm}^0, \quad (33)$$

is exponentially attracting. Condition (26), that the reduced system has attracting sliding ( $h_{\text{rd},x} f_+^0 < 0 < h_{\text{rd},x} f_-^0$ ) is equivalent to the conditions

$$h_x^0 f_+^0 < h_y^0 y_{\text{sl},+}, \quad h_x^0 f_-^0 > h_y^0 y_{\text{sl},-} \quad (34)$$

on  $y_{\text{sl},\pm}$ . Assuming that

$$h_x^0 (f_+^0 - f_-^0) \neq 0, \quad (35)$$

the full truncated system has

– attractive sliding if  $h_x^0 f_+^0 < h_x^0 f_-^0$  for  $(x, y) \in \partial \mathcal{H}_0$  where  $y$  satisfies

$$h_x^0 f_+^0 < h_y^0 y < h_x^0 f_-^0 \quad (36)$$

– repelling sliding if  $h_x^0 f_+^0 > h_x^0 f_-^0$  for  $(x, y) \in \partial \mathcal{H}_0$  where  $y$  satisfies

$$h_x^0 f_+^0 > h_y^0 y > h_x^0 f_-^0 \quad (37)$$

(conditions are independent of the  $x$  component). The boundary of these sliding regions are the affine tangent subspaces

$$\mathcal{H}_{\text{tan},\pm} = \{(x, y) \in \partial \mathcal{H}_0 : h_y^0 y = h_x^0 f_{\pm}^0\} \quad (38)$$

(these are subspaces of the switching subspace  $\partial \mathcal{H}_0$ ). The locations of the subspaces  $\mathcal{M}_{\text{slow},\pm}$  and  $\mathcal{H}_{\text{tan},\pm}$  relative to each other will determine the possible signatures of the dynamics (sequencing switches and sliding along trajectories) in the full truncated system in Sect. 4.

For sliding in the full truncated system (32) the factor  $\alpha$  for the Filippov convention (2) simplifies to (assuming  $h_x^0 f_+^0 \neq h_x^0 f_-^0$  as in (35))

$$\alpha = \frac{h_x^0 f_+^0 - h_y^0 y}{h_x^0 (f_+^0 - f_-^0)}, \quad (39)$$

which depends on  $y$ . Thus, the sliding vector field  $F_s$  of the truncated full system (32) is an affine flow:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = F_s = b_s + A_s \begin{bmatrix} x \\ y \end{bmatrix}, \text{ where}$$

$$b_s = \begin{bmatrix} f_+^0 \\ -g_x^0 f_+^0 \end{bmatrix} - \frac{h_x^0 f_+^0}{h_x^0 (f_-^0 - f_+^0)} \begin{bmatrix} f_-^0 - f_+^0 \\ -g_x^0 (f_-^0 - f_+^0) \end{bmatrix}, \text{ and}$$

$$A_s = \begin{bmatrix} 0 & \frac{(f_-^0 - f_+^0)h_y^0}{h_x^0(f_-^0 - f_+^0)} \\ 0 & g_y^0 - \frac{g_x^0(f_-^0 - f_+^0)h_y^0}{h_x^0(f_-^0 - f_+^0)} \end{bmatrix}. \quad (40)$$

We observe that the matrix  $A_s$  in the linear term of  $F_s$  has the  $n$ -dimensional nullspace  $\{(x, y) : y = 0, x \text{ arbitrary}\}$ . Furthermore, the dynamics of  $F_s$  does not need to be linearly stable even though  $g_y^0$  is assumed to generate stable dynamics because of the non-small rank-1 perturbation  $g_x^0(f_-^0 - f_+^0)h_y^0/(h_x^0(f_-^0 - f_+^0))$ . The two conditions, (26) and (35),

$$h_x^0(f_-^0 - f_+^0) \neq 0, \quad h_{rd,x}^0(f_-^0 - f_+^0) > 0$$

ensure that the U-singularity [2] occurs neither in the sliding vector field  $f_{rd,s}^0$  of the reduced truncated system, nor in the affine sliding vector field  $F_s$  of the full truncated system.

#### 4 Classification of switching signatures

The different possible switching signatures can be classified by considering the relative locations of the attracting subspace  $\mathcal{M}_{slow,\pm}$  for each of the vector fields  $F_{\pm}$  near the switching subspace  $\partial\mathcal{H}_0$  and the tangency subspaces  $\mathcal{H}_{tan,\pm}$ , which determine where sliding regions in  $\partial\mathcal{H}_0$  start or end. Conveniently, all subspaces are determined by the projection  $h_y^0$  of the  $y$ -component of the phase space variable, such that for each  $(x, y)$  we may consider its linear projection into the plane

$$P : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^2 \quad \text{with} \\ [P_1(x, y), P_2(x, y)] = \left[ h_{rd,x}^0 x - h_y^0 \left[ g_y^0 \right]^{-1} y, h_y^0 y \right].$$

In projection  $P$  the switching subspace  $\partial\mathcal{H}_0$  is the vertical axis

$$\partial\mathcal{H}_0 = \{(x, y) : P_1(x, y) = 0\},$$

such that the vector fields  $F_{\pm}$  are applicable in the half spaces projecting onto the positive and negative half-planes

$$\mathcal{H}_{\pm} = \{(x, y) : \pm P_1(x, y) > 0\}.$$

The projections of the tangent subspaces for sliding in  $\partial\mathcal{H}_0$  are the points

$$(0, T_{\pm}), \quad \text{where } T_{\pm} = h_x^0 f_{\pm}^0, \text{ such that} \\ \mathcal{H}_{tan,\pm} = \{(x, y) : P_2(x, y) = T_{\pm}\}.$$

The projections of the attracting invariant subspaces  $\mathcal{M}_{slow,\pm}$  of  $F_{\pm}$  for the regions  $\mathcal{H}_{\pm}$  are the horizontal levels

$$\mathcal{M}_{P,slow,\pm} = \{(p_1, p_2) : \pm p_1 > 0, p_2 = S_{\pm}\} \text{ where}$$

$$S_{\pm} = h_{y,sl,\pm}^0, \text{ such that}$$

$$\mathcal{M}_{slow,\pm} = \{(x, y) : \pm P_1(x, y) > 0, P_2(x, y) = S_{\pm}\}.$$

In projection  $P = (p_1, p_2)$ , the nullclines for the horizontal component of motion in half plane  $\{\pm p_1 > 0\}$  are the horizontal levels  $T_{\pm}$ :  $\{\pm p_1 > 0, p_2 = h_y^0 f_{\pm}^0\}$ . The vertical component of motion is attracted toward  $\mathcal{M}_{P,slow,\pm}$  in each half-plane. If  $y$  is scalar (such that we have only one fast variable in the original system (3), (4)), the line  $\mathcal{M}_{P,slow,\pm}$  acts as a nullcline for vertical motion. For  $\dim y = m > 1$  this may not be the case such that the vertical component of arrows in Fig. 1 is for illustration only. By our assumption (26) that the reduced flow has attractive sliding the levels  $T_{\pm}$  and levels  $S_{\pm}$  must satisfy

$$T_- > S_-, \quad S_+ > T_+, \quad (41)$$

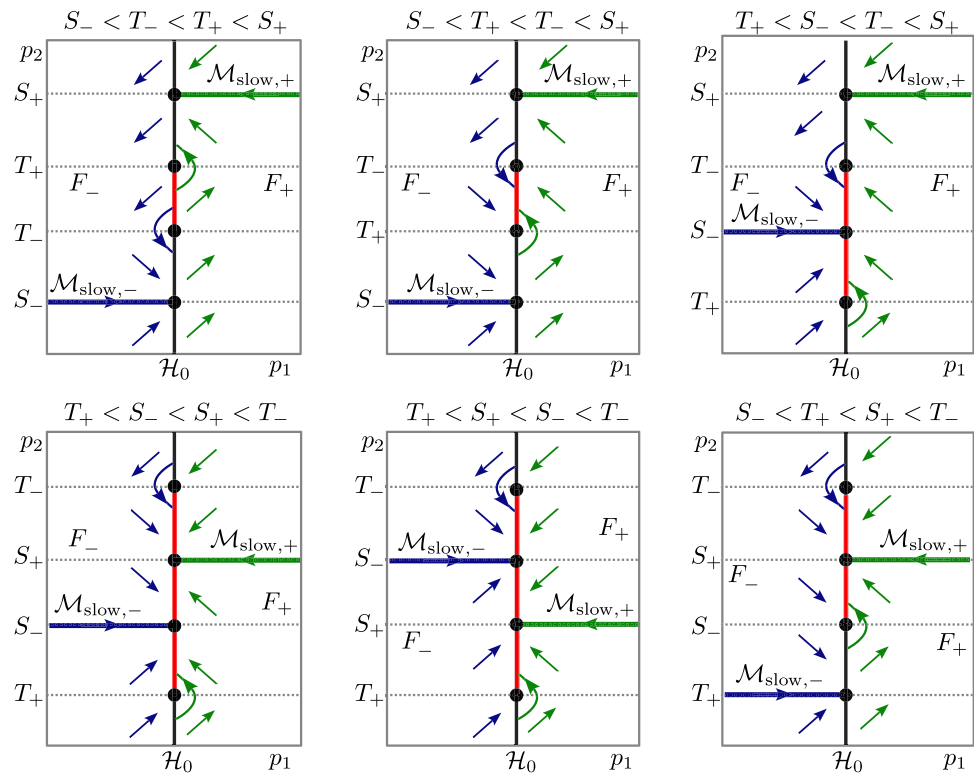
resulting in the six possible arrangements of  $S_{\pm}$  and  $T_{\pm}$  illustrated in Fig. 1 and classified in Table 1.

The cases  $T_+ < S_- < T_- < S_+$  and  $S_- < T_+ < S_+ < T_-$  are equivalent when interchanging the symbols  $+$  and  $-$  and the vector fields  $F_+$  and  $F_-$ . Thus, overall, we have five qualitatively different scenarios up to change of subscripts  $+$  and  $-$ . As Fig. 1 shows, the sliding region along the vertical axis  $\{p_1 = 0\}$  always occurs between points  $T_+$  and  $T_-$  (and sliding is attracting if  $T_+ < T_-$ ). The right column in Table 1 shows switching transitions that are possible in the long-term dynamics near  $\partial\mathcal{H}_0 \cap (\mathcal{M}_{P,slow,+} \cup \mathcal{M}_{P,slow,-})$ . We observe that there is an upper limit for the time the flow can follow  $F_{\pm}$  such that every period of following  $F_{\pm}$  (indicated by symbol  $\pm$  in Table 1) must be followed by a switch, either to  $F_{\mp}$  (symbol  $\mp$  in Table 1), or to  $F_s$  (symbol  $s$  in Table 1).

Scenario  $S_- < T_- < T_+ < S_+$  (top row in Table 1, see Fig. 1) As the sliding set is repelling between the levels  $T_+$  and  $T_-$  there exists a repelling sliding set. This implies that any trajectory will follow  $F_+$  and  $F_-$  alternately with infinitely many switches from  $F_-$  to  $F_+$  between  $S_-$  and  $T_-$ , and from  $F_+$  to  $F_-$  between  $T_+$  and  $S_+$ . In Fig. 1 this is schematically indicated by the arrows pointing away from the vertical line depicting the switching manifold. These arrows indicate the directions of the vector fields  $F_{\pm}$ .

Scenario  $S_- < T_+ < T_- < S_+$  (row 2 in Table 1, see Fig. 1) In the region between  $T_+$  and  $T_-$  there exists an attracting sliding set. This is schematically indicated by the arrows pointing towards the vertical line depicting the switching manifold. Every transition between flows is possible in

**Fig. 1** Projection of phase portraits into plane by projection  $(p_1, p_2) = P(x, y) = (h_{rd,x}^0 x - h_y^0 [g_y^0]^{-1} y, h_y^0 y)$  for cases listed in Table 1. Note that the sketched vector fields are not in a phase plane. The sign of the horizontal component of all arrows is correctly reflected in the sketch. The vertical component is only illustrative (indicating attractivity of  $\mathcal{M}_{slow,\pm}$  for  $F_{\pm}$ ). Sections 5 and 6 study the case in the top-left panel,  $S_- < T_- < T_+ < S_+$



**Table 1** Possible transitions of dynamics near switching subspace  $\partial\mathcal{H}_0$  of truncated full systems (32).

| Vertical ordering       | Sliding    | Possible long-term switching |
|-------------------------|------------|------------------------------|
| $S_- < T_- < T_+ < S_+$ | Repelling  | $+-, -+$                     |
| $S_- < T_+ < T_- < S_+$ | Attracting | $+-, -+, s, s+, s-, +s, -s$  |
| $T_+ < S_- < T_- < S_+$ | Attracting | $-s, s-, s$                  |
| $T_+ < S_- < S_+ < T_-$ | Attracting | $s$                          |
| $T_+ < S_+ < S_- < T_-$ | Attracting | $s$                          |
| $S_- < T_+ < S_+ < T_-$ | Attracting | $+s, s+, s$                  |

Left column: ordering of  $T_{\pm}$  for tangent set  $\mathcal{H}_{tan,\pm}$  and of  $S_{\pm}$  for attracting subspace  $\mathcal{M}_{P,slow,\pm}$  (see vertical ordering of  $S_{\pm}$  and  $T_{\pm}$  in Fig. 1). Middle column: nature of sliding on sliding segment (attracting or repelling) according to (36) and (37). Right column: switches (between  $\mathcal{H}_+$ ,  $\mathcal{H}_-$  and sliding that are possible in the long term: symbol  $+-$  stands for crossing from  $\mathcal{H}_+$  to  $\mathcal{H}_-$ ,  $-+$  vice versa,  $s_{\pm}$  for leaving sliding toward  $\mathcal{H}_{\pm}$ ,  $\pm s$  for entering sliding from  $\mathcal{H}_{\pm}$ , single  $s$  for long-term sliding. See also Fig. 1 for illustration

the long run (apart from permanently following  $F_+$  or permanently following  $F_-$  not being possible). Which sequence actually occur depends on the coefficients of the problem.

Scenario  $T_+ < S_- < T_- < S_+$  (row 3 in Table 1, see Fig. 1) In this scenario the flow will not visit  $\mathcal{H}_+$  (thus, not follow  $F_+$  in the long term). The flow may permanently follow  $F_s$  (sliding), or show infinitely many excursions into  $\mathcal{H}_-$  (following  $F_-$ ). The case  $S_- < T_+ < S_+ < T_-$  (row 6 in Table 1) is a mirror image of this scenario when interchanging subscripts  $+$  and  $-$ . There the flow can have infinitely many excursions into  $\mathcal{H}_+$ , but will not visit  $\mathcal{H}_-$ .

Scenarios  $T_+ < S_- < S_+ < T_-$  and  $T_+ < S_+ < S_- < T_-$  (rows 4 and 5 in Table 1, see Fig. 1) As the attracting

subspaces  $\mathcal{M}_{slow,\pm}$  both intersect the switching manifold in the attracting sliding set, the flow permanently follows sliding ( $F_s$ ).

The quantities  $T_{\pm}$  and  $S_{\pm}$  are constants in the rescaled system (32) such that the affine subspaces  $\{(x, y) : P(x, y) = (0, S_{\pm})\}$  and  $\{(x, y) : P(x, y) = (0, T_{\pm})\}$  are by construction parallel to each other after zooming in to obtain (32). As  $S_{\pm}$  and  $T_{\pm}$  depend on  $x_0$  these subspaces are tangent spaces to the following nonlinear sets. In the original coordinates the coefficients defining the spaces depend on the base point  $x_0$ , such that in special points the spaces are equal (whenever the underlying quantities  $T_{\pm}(x_0, y_c(x_0))$  and  $S_{\pm}(x_0, y_c(x_0))$  coincide.

All points  $x_0$  at which the vertical ordering of  $S_{\pm}$  and  $T_{\pm}$  changes correspond to higher-codimension sub-manifolds of the switching boundary  $\mathcal{H}_0$  intersected with the critical manifold  $\mathcal{M}_0$ . Whenever a sliding trajectory of the reduced system is near one of these submanifolds a degeneracy will occur in one of the coefficients in the analysis such as in Sect. 5. Thus, near these points separate analysis is required. For example, [18] studied the case where  $S_+ = T_+$ ,  $S_- < T_-$ , where one of the reduced vector fields is grazing (here  $h_{rd} > 0$ ), and the other one is pointing toward the switching surface.

## 5 General return map for repelling sliding case

In the remaining part of the paper we will focus on the case where the full truncated system has repelling sliding (while the reduced truncated sliding has attracting sliding by assumption). So we have the relations

$$h_{y, y_{sl}, -}^0 < h_x^0 f_-^0 < h_x^0 f_+^0 < h_{y, y_{sl}, +}^0. \quad (42)$$

Since the sliding segment is repelling, initial conditions starting in the sliding segment do not have a unique forward trajectory. However, each point in the repelling sliding segment has admissible non-sliding forward trajectories. For this reason we only consider non-sliding trajectories of the full truncated system (32) in this section.

We will start with the following Lemma, establishing a return map from the half subspace of the switching subspace

$$\mathcal{H}_0^+ := \{(x, y) \in \partial\mathcal{H}_0 : h_y^0 y \geq h_x^0 f_+^0\}$$

back into itself. The projection  $P$  of the set  $\mathcal{H}_0^+$  is the part of the vertical axis above point  $T_+$  in Fig. 1.

**Lemma 4** (Continuous switching for the repelling sliding case) *Let  $(x_0, y_0) \in \mathcal{H}_0^+$ . There exists a trajectory  $(X_-^t(x_0), Y_-^t(y_0))$  following  $F_-$  and a time  $t_1 > 0$ , such that  $(X_-^{t_1}(x_0), Y_-^{t_1}(y_0)) \in \mathcal{H}_-$  for all  $t \in (0, t_1)$ ,  $(x_1, y_1) = (X_-^{t_1}(x_0), Y_-^{t_1}(y_0))$  is in the switching subspace  $\partial\mathcal{H}_0$  and satisfies  $h_y^0 y_1 \leq h_x^0 f_-^0$ .*

*There also exists a trajectory  $(X_+^t(x_1), Y_+^t(y_1))$  following  $F_+$  and a time  $t_2 > 0$ , such that  $(X_+^{t_2}(x_1), Y_+^{t_2}(y_1)) \in \mathcal{H}_+$  for all  $t \in (0, t_2)$ , and  $(x_2, y_2) = (X_+^{t_2}(x_1), Y_+^{t_2}(y_1))$  is back in  $\mathcal{H}_0^+$ . An explicit expression for  $(x_2, y_2)$  as a function of the initial point  $(x_0, y_0)$  and times  $t_1$  and  $t_2$  is*

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_0 + f_-^0 t_1 + f_+^0 t_2 \\ e^{(t_1+t_2)g_y^0} y_0 + [e^{t_2 g_y^0} - e^{(t_1+t_2)g_y^0}] y_{sl, -} + [I - e^{t_2 g_y^0}] y_{sl, +} \end{bmatrix}$$

where  $t_1$  and  $t_2$  are functions of  $y_0$ , but independent of  $x_0$ .

**Proof** The truncated flows for  $F_{\pm}$ , given in (32), are decoupled in  $x$  and  $y$ , affine and have the explicit formula

$$X_{\pm}^t(x) = x + f_{\pm}^0 t, \quad Y_{\pm}^t(y) = y_{sl, \pm} + e^{t g_y^0} (y - y_{sl, \pm}). \quad (43)$$

The switching function of (32) along such a trajectory is the function

$$h_{\pm}(t; x, y) = h_{rd, x}^0 [x + f_{\pm}^0 t] + h_y^0 [g_y^0]^{-1} [y_{sl, \pm} + e^{t g_y^0} (y - y_{sl, \pm})]. \quad (44)$$

For  $(x, y) \in \partial\mathcal{H}_0$ , we have  $h_{\pm}(0; x, y) = 0$  by definition of  $\partial\mathcal{H}_0$ , such that (44) simplifies to

$$h_{\pm}(t; y) = h_{rd, x}^0 f_{\pm}^0 t + h_y^0 [g_y^0]^{-1} [e^{t g_y^0} - I] [y - y_{sl, \pm}]. \quad (45)$$

So,  $h_{\pm}$  is independent of  $x$ . Its time derivative at  $t = 0$  in some  $(x, y)$  and the limit for  $t \rightarrow \infty$ , when following  $F_{\pm}$  are

$$\begin{aligned} \partial_t h_{\pm}(0; y) &= h_x^0 f_{\pm}^0 - h_y^0 y, \\ \lim_{t \rightarrow \infty} h_{\pm}(t; y)/t &= h_x^0 f_{\pm}^0 - h_y^0 y_{sl, \pm}, \end{aligned} \quad (46)$$

because  $g_y^0$  is a matrix with eigenvalues with negative real parts.

As  $h_y^0 y_0 > h_x^0 f_+^0 > h_x^0 f_-^0$ , vector field  $F_-$  points into  $\mathcal{H}_-$  at  $(x_0, y_0)$ , while vector field  $F_+$  points into  $\mathcal{H}_-$  or is tangent to  $\partial\mathcal{H}_0$ , such that we can follow  $F_-$ .

Using (46) we observe that  $h_-(0; y_0) = 0$ ,  $\partial_t h_-(0; y_0) = h_x^0 f_-^0 - h_y^0 y_0 < 0$  by assumption that  $(x_0, y_0) \in \mathcal{H}_0^+$ , and that  $\lim_{t \rightarrow \infty} h_-(t; y_0)/t = h_x^0 f_-^0 - h_y^0 y_{sl, -} > 0$  by (42). Thus,  $t \mapsto h_-(t; y_0)$  must reach zero for some  $t_1 > 0$ . Let us call  $(x_1, y_1) = (X_-^{t_1}(x_0), Y_-^{t_1}(y_0))$  for this reaching time  $t_1$ .

At point  $(x_1, y_1) \in \partial\mathcal{H}_0$  we have  $\partial_t h_-(t_1; y_0) = \partial_t h_-(0; y_1) = h_x^0 f_-^0 - h_y^0 y_1 \geq 0$ . Since  $h_x^0 f_+^0 - h_x^0 f_-^0$  (see (42)), we have  $h_x^0 f_+^0 - h_y^0 y_1 > 0$  such that at  $(x_1, y_1)$  vector field  $F_+$  points into  $\mathcal{H}_+$ , while  $F_-$  points into  $\mathcal{H}_+$  or is tangent to  $\partial\mathcal{H}_0$ , such that we can follow  $F_+$ . We have  $h_+(0; y_1) = 0$ ,  $\partial_t h_+(0; y_1) = h_x^0 f_+^0 - h_y^0 y_1 > 0$  and  $\lim_{t \rightarrow \infty} h_+(t; y_1)/t = [h_x^0 f_+^0 - h_y^0 y_{sl, +}] < 0$  according to (42). Thus,  $t \mapsto h_+(t; y_1)$  must reach zero for some  $t_2 > 0$ , and we call  $(x_2, y_2) = (X_+^{t_2}(x_1), Y_+^{t_2}(y_1))$  for this reaching time  $t_2$ . The resulting  $(x_2, y_2)$  satisfies  $(x_2, y_2) \in \mathcal{H}_0^+$  as  $\partial_t h_+(t_2; y_1) = \partial_t h_+(0; y_2) = h_x^0 f_+^0 - h_y^0 y_2 \leq 0$ .

Combining the expressions for  $X_{\pm}^t$  and  $Y_{\pm}^t$ ,  $x_2 = X_+^{t_2} \circ X_-^{t_1}(x_0)$  and  $y_2 = Y_+^{t_2} \circ Y_-^{t_1}(y_0)$ , we express point  $(x_2, y_2)$  as

$$\begin{aligned} x_2 &= x_0 + f_-^0 t_1 + f_+^0 t_2, \\ y_2 &= e^{(t_1+t_2)g_y^0} y_0 + (e^{t_2 g_y^0} - e^{(t_1+t_2)g_y^0}) y_{sl,-} \\ &\quad + (I - e^{t_2 g_y^0}) y_{sl,+}. \end{aligned}$$

By construction, we had that  $t_1$  and  $y_1$  are function of  $y_0$ , but not of  $x_0$ , such that  $t_2$  as a function of  $y_1$  (but not of  $x_1$ ) is a function of  $y_0$ , but not of  $x_0$ .  $\square$

Lemma 4 permits us to define a return map, which has a skew-product form (the  $x$  component is driven by a map for the  $y$  component)

$$R: \mathcal{H}_0^+ \rightarrow \mathcal{H}_0^+, R(x, y) = (R_x(x, y), R_y(y))$$

as constructed in Lemma 4.

We denote the  $x$  and  $y$  components of  $R$  by  $R_x$  and  $R_y$  respectively. Furthermore, we denote the travel times constructed as part of the construction in Lemma 4 by

$$\begin{aligned} t_{\pm}: \mathcal{H}_{0,y}^+ &:= \{y \in \mathbb{R}^m : h_y^0 y \geq h_x^0 f_+^0\} \rightarrow (0, \infty), \\ t_-(y) &:= t_1, \quad t_+(y) := t_2 \quad \text{as in Lemma 4,} \end{aligned}$$

recalling that (since  $h_{rd,x}^0 \neq 0$ ) we can find for every  $y \in \mathbb{R}^m$  a  $x \in \mathbb{R}^n$  such that  $(x, y) \in \partial \mathcal{H}_0$ . If we restrict  $y$  to the half space  $\{y \in \mathbb{R}^m : h_y^0 y \geq 0\}$ , then  $(x, y) \in \mathcal{H}_0^+$ . Furthermore, the travel time will be independent of the  $x$  we choose, making the definition of  $t_{\pm}$  well defined. Using  $t_{\pm}$  we can express  $R_x$  as

$$R_x(x, y) = x + f_-^0 t_-(y) + f_+^0 t_+(y). \quad (47)$$

**Corollary 1** (Eventual boundedness of travel times and solutions) *There exists a closed bounded subset  $\mathcal{H}_{\infty,y}^+ \subseteq \mathcal{H}_{0,y}^+$  such that for every  $y \in \mathcal{H}_{\infty,y}^+$  the orbit  $R_y^\ell(y)$  will eventually stay in  $\mathcal{H}_{\infty,y}^+$ . A bound for the iterate  $\ell_{\min}(y)$  with  $R^\ell(y) \in \mathcal{H}_{\infty,y}^+$  for all  $\ell \geq \ell_{\min}(y)$  depends only on the norm of  $y$ , such that we can write  $\ell_{\min}(|y|)$ .*

*There exist bounds  $0 < t_{\text{low}} < t_{\text{up}}$  such  $t_{\pm}(y)$  satisfy  $t_{\text{low}} \leq t_{\pm}(y) \leq t_{\text{up}}$  for all  $y \in \mathcal{H}_{\infty,y}^+$ .*

**Proof** In the construction of  $t_{\pm}(y)$  in Lemma 4 the lower bound  $t_{\text{low}}$  is implied by Assumption (37) for the scenario we consider. There exists a regular matrix  $B \in \mathbb{R}^{m \times m}$  such that  $e^{t g_y^0}$  for all  $t \geq t_{\text{low}} > 0$  is a contraction in the norm  $|\cdot|_B$  induced by the inner product  $y^T B^T B y$ . In this norm the flows  $Y_{\pm}^t$  map into a bounded ball in the norm  $|\cdot|_B$ . A possible choice for the set  $\mathcal{H}_{\infty,y}^+$  is the closure of this bounded ball. The time it takes for the flow to map into  $\mathcal{H}_{\infty,y}^+$  depends only on the norm of the initial value  $y$ . For  $y \in \mathcal{H}_{\infty,y}^+$

and  $t \geq t_{\text{low}}$  the term  $h_y^0 [g_y^0]^{-1} [e^{t g_y^0} - I] [y - y_{sl,\pm}]$  in (45) is bounded, such that the limit for large  $t$  is uniform for  $y \in \mathcal{H}_{\infty,y}^+$ , providing a uniform upper bound  $t_{\text{up}}$  for  $t_{\pm}(y)$ .  $\square$

For an orbit  $R^\ell(x, y)$  the  $x$  component satisfies (denoting the  $x$  component of  $R^\ell(x, y)$  by  $R_x^\ell(x, y)$ )

$$\begin{aligned} R_x^\ell(x, y) &= x + f_+^0 T_{\ell,+}(y) + f_-^0 T_{\ell,-}(y), \quad \text{where} \\ T_{\ell,\pm}(y) &= \sum_{i=0}^{\ell-1} t_{\pm}(R_y^i(y)), \end{aligned} \quad (48)$$

defining  $T_\ell(y) = T_{\ell,+}(y)T_{\ell,-}(t)$  as the orbit's travel time. For  $y \in \mathcal{H}_{\infty,y}^+$  Corollary 1 implies that  $R_x^\ell(x, y)$  drifts with speed proportional to  $\ell$ , where the times  $T_{\ell,\pm}$  are both bounded by  $[\ell t_{\text{low}}, \ell t_{\text{up}}]$ , in the affine plane spanned by  $f_{\pm}^0$ , rooted at  $x$ .  $\square$

**Lemma 5** (Long-term drift in slow variable) *Let  $(x, y) \in \mathcal{H}_0^+$  be arbitrary. For large  $\ell$  the orbit  $R^\ell(x, y)$  satisfies*

$$\frac{R_x^\ell(x, y) - x}{T_\ell(y)} = f_{rd,s}^0 + O(1/\ell). \quad (49)$$

*The bounding constant in  $O(1/\ell)$  depends only on the norm  $|y|$ .*

The vector  $f_{rd,s}^0$ , given in (29), is the right-hand side of the truncated reduced sliding flow.

**Proof** Define  $\alpha_\ell(y) = T_{\ell,-}(y)/T_\ell(y)$ . Then

$$\frac{R_x^\ell(x, y) - x}{T_\ell(y)} = f_-^0 \alpha_\ell(y) + f_+^0 (1 - \alpha_\ell(y)). \quad (50)$$

Since  $R^\ell(x, y)$  and  $(x, y)$  lie in the switching subspace  $\partial \mathcal{H}_0$ , they satisfy

$$\begin{aligned} 0 &= h_{rd,x}^0 \left[ x + f_+^0 T_{\ell,+}(y) + f_-^0 T_{\ell,-}(y) \right] \\ &\quad - h_y^0 [g_y^0]^{-1} R_y^\ell(y), \\ 0 &= h_{rd,x}^0 x - h_y^0 [g_y^0]^{-1} y. \end{aligned}$$

Subtracting the two identities and dividing them by  $T_\ell(y)$ , we get

$$\begin{aligned} 0 &= h_{rd,x}^0 f_+^0 (1 - \alpha_\ell(y)) + h_{rd,x}^0 f_-^0 \alpha_\ell(y) \\ &\quad - \frac{1}{T_\ell(y)} h_y^0 [g_y^0]^{-1} [R_y^\ell(y) - y]. \end{aligned}$$

We can re-arrange this identity for  $\alpha_\ell(y)$ , obtaining

$$\alpha_\ell(y) = \frac{h_{rd,x}^0 f_+^0}{h_{rd,x}^0 (f_+^0 - f_-^0)}$$

$$-\frac{1}{T_\ell(y)} \frac{h_y^0 [g_y^0]^{-1} [R_y^\ell(y) - y]}{h_{rd,x}^0 (f_+^0 - f_-^0)}.$$

In the second term the denominator  $h_{rd,x}^0 f_+^0 - h_{rd,x}^0 f_-^0$  is a negative (in particular, non-zero) constant, the term  $R_y^\ell(y)$  is bounded by  $y_{bd} = \sup\{|y| : y \in \mathcal{H}_{\infty,y}^+\}$ . The denominator  $T_\ell(y)$  can be bounded by  $T_\ell(y) \geq \ell t_{low}$ , such that

$$|\alpha_\ell(y) - \alpha_{rd}^0| \leq \frac{1}{\ell} \frac{\|h_y^0 [g_y^0]^{-1}\| (y_{bd} + |y|)}{t_{low} |h_{rd,x}^0 (f_+^0 - f_-^0)|}. \quad (51)$$

This means that replacing  $\alpha_\ell(y)$  by  $\alpha_{rd}^0$  in (50) results in an error of order  $1/\ell$  with a factor as shown on the right in (51), multiplied by  $|f_+^0| + |f_-^0|$ . This shows the claim of the lemma.  $\square$

The arguments in the proof of Lemma 4 also imply the following that from every initial point a trajectory reaches the domain  $\mathcal{H}_0^+$  of the map  $R$ .

**Corollary 2** (Reachability of crossing set) *For every initial condition  $(x, y)$  outside of repelling sliding segment*

$$\mathcal{H}_{0,su} = \{(x, y) \in \mathcal{H}_0 : h_y^0 y \in [h_x^0 f_-^0, h_x^0 f_+^0]\}$$

*the trajectory of (32) will reach  $\mathcal{H}_0^+$  after a finite time  $t_{ini}(y)$ . For  $(x, y)$  inside the repelling sliding segment  $\mathcal{H}_{0,su}$  (where the forward trajectory is not unique) any admissible non-sliding trajectory leaves  $\mathcal{H}_{0,s}$  and also reaches  $\mathcal{H}_0^+$  after finite time  $t_{ini}(y)$ . The time  $t_{ini}(y)$  has a uniform upper bound  $t_{bd}$  for each ball  $\{y : |y| \leq y_{bd}\}$ .*

Corollary 2 and Lemma 5 permit us to make the following statement about the quality of the approximation of a trajectory of the truncated full system (32) to the sliding trajectory of the truncated reduced system (27).

**Theorem 6** (Asymptotic sliding approximation) *Assume that the truncated full system (32) satisfies the conditions (42) for repelling sliding. Let  $(x(t), y(t))$  be a trajectory of (32) starting from  $(x(0), y(0)) = (x_0, y_0)$  and consisting only of non-sliding segments. Then for large times  $t$*

$$\frac{x(t) - x(0)}{t} = f_{rd,s}^0 + O(1/t). \quad (52)$$

*The bound in  $O(1/t)$  only depends on  $|y|$ .*

Rescaling coordinates back into original  $x, y$  and time scale results in the statement of Theorem 3.

## 6 Case $m = 1$ , one-dimensional fast variable with repelling sliding

### 6.1 Periodic switchings

We will show that a natural situation of the dynamics of Scenario  $S_- < T_- < T_+ < S_+$  when the fast variable is one-dimensional implies the existence of a stable fixed point of mapping  $R$ . This implies further that the equality given in Theorem 6 is exact, that is:

$$\frac{x(t) - x(0)}{t} = f_{rd,s}^0.$$

W.l.o.g., we assume that switchings depend on  $x^1$  state only, where  $x = [x^1, x^2, \dots, x^n]$ , and also on  $y$  state. Hence  $h_{rd,x}^0 = [1 \ 0, \dots, 0]$  and  $h_y^0 = 1$ . By rescaling  $\varepsilon$  in the initial fast variable equation (3), (4), we may set  $g_y^0 = -1$ .

The switching surface is then given by  $\partial\mathcal{H}_0 = \{x^1 + y = 0\}$ . We suppose that we start at some point  $(x_0^1, *, y_0)$  where the “\*” symbol indicates remaining  $n - 2$  initial values of vector field  $F_-$ . After some  $t_1 > 0$  we reach the switching surface again at some point  $(x_1^1, *, y_1)$ . From  $(x_1^1, *, y_1)$  we follow vector field  $F_+$  for some  $t_2 > 0$  until the switching surface is reached again at point  $(x_2^1, *, y_2)$ . Note that this scenario is ensured under conditions stated in Lemma 4. We get the following set of equations which define the evolution of our system

$$\begin{aligned} x(t_1) = x_1 = x_0 + f_-^0 t_1, \\ y(t_1) = y_1 = e^{-t_1} (y_0 + g_x^0 f_-^0) - g_x^0 f_-^0, \end{aligned} \quad (53)$$

$$\begin{aligned} x(t_2) = x_2 = x_1 + f_+^0 t_2, \\ y(t_2) = y_2 = e^{-t_2} (y_1 + g_x^0 f_+^0) - g_x^0 f_+^0, \end{aligned} \quad (54)$$

$$\begin{aligned} x_0^1 + y_0 = 0, \\ x_1^1 + y_1 = 0, \quad x_2^1 + y_2 = 0. \end{aligned} \quad (55)$$

Let us now suppose that  $x_0^1 = x_2^1$  and  $y_2 = y_0$ . It is natural to ask whether such a scenario is possible, and if so, how generic it is, and finally what are the consequences of such a motion. In what follows, such a motion will be termed a period-one orbit. Rearranging and combining equations (53)–(55) give the conditions for the existence of such an orbit, which can be expressed as:

$$0 = x_0^1 + y_0 = 0, \quad (56)$$

$$0 = x_0^1 + h_{rd,x}^0 f_-^0 t_1 + e^{-t_1} (y_0 + g_x^0 f_-^0) - g_x^0 f_-^0, \quad (57)$$

$$0 = t_1 + t_2 \frac{h_{rd,x}^0 f_+^0}{h_{rd,x}^0 f_-^0}, \quad (58)$$

$$y_0 + g_x^0 f_+^0 = e^{-(t_1+t_2)} (y_0 + g_x^0 f_-^0) - e^{-t_2} g_x^0 (f_-^0 - f_+^0). \quad (59)$$

Conditions (56) and (57) imply that points  $(x_0, y_0)$  and  $(x_1, y_1)$  lie on the switching manifold, whereas conditions (58) and (59) result from periodicity conditions  $x_0^1 = x_2^1$  and  $y_0 = y_2$ . It can be shown that under the conditions of Lemma 4, matrix  $-g_x^0$  may be found which satisfies the above equations for  $t_1$  ( $t_2$ ) and  $y_0$  for an open set of the matrix values and hence a period-1 orbit is a generic case. Now, what are the consequence of such dynamics? Consider equation (58), which relates the times of evolution of flows generated by vector fields  $F_{\pm}$ . We have that

$$\frac{t_1}{t_1 + t_2} = -\frac{h_{rd,x}^0 f_+^0}{h_{rd,x}^0 (f_-^0 - f_+^0)} = \alpha, \quad (60)$$

where  $\alpha$  is given by equation (28), and  $\alpha$  represents the same ratio of times as the one in equation (60) above, but  $\alpha$  gives it for the reduced sliding flow. Hence, if there is a stable period-1 solution in the full flow then the dynamics of the full flow corresponds to the dynamics of the sliding flow of the reduced system in that the drift along the slow dynamics is equivalent to the motion of the sliding flow in the reduced system, if we consider projecting the full flow onto the sliding flow. Or putting it yet another way, the equality given in Theorem 6 is exact.

*The existence of period-1 cycles* We focus further on equations (56)–(59). Consider now equation (58) which defines time  $t_1$  as a function of periodic point  $y_0$ . We have:

$$-y_0 + h_{rd,x}^0 f_-^0 t_1 + e^{-t_1} (y_0 + g_x^0 f_-^0) - g_x^0 f_-^0 = 0. \quad (61)$$

Let us suppose that  $t_1$  is small, so we may expand expression (61) in  $t_1$  to leading order. We get:

$$-y_0 + h_{rd,x}^0 f_-^0 t_1 + (1 - t_1)(y_0 + g_x^0 f_-^0) - g_x^0 f_-^0 = 0,$$

which simplifies to:

$$t_1(h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0) = 0. \quad (62)$$

Note that for (62) to be equal to 0, we require

$$h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0 = h_x^0 f_-^0 - y_0 = 0,$$

which is satisfied at the boundary of the sliding set of the full system. Hence, the existence of period-1 cycles is bounded, for small times  $t_1$ , by the existence of the sliding set in the full system.

Another important value of  $y_0$  is  $y_0 = -g_x^0 f_-^0$ . We note that on the set  $\{y_0 = -g_x^0 f_-^0\}$  we have  $\dot{y} = 0$ , and hence this set defines the slow manifold of the full system, which, in turn, defines another boundary for the existence of period-1 cycles.

## 6.2 Stable period-1 cycles

Clearly, if the system exhibits stable period-1 cycles as defined in the previous section then the singular perturbation of the sliding flow does not significantly alter system dynamics. Hence, it is of our interest to determine the set of parameters for which such a scenario takes place. To this aim consider a map from some  $(x_0, *, y_0) \in \mathcal{H}_0^+$  back to  $\mathcal{H}_0^+$  under the composition of flows  $(X_+^{t_2}(x_1), Y_+^{t_2}(y_1)) \circ (X_-^{t_1}(x_0), Y_-^{t_1}(y_0))$ . We drop the superscript “1” in the  $x_0^1$  component for the clarity of the notation. We know that such a map exists under the conditions of Lemma 4. We have that

$$y_2 = e^{-(t_1+t_2)}(y_0 + g_x^0 f_-^0) - e^{-t_2} g_x^0 (f_-^0 - f_+^0) - g_x^0 f_+^0. \quad (63)$$

Obviously  $t_1$  and  $t_2$  are functions of  $y_0$ . We may linearize above map (63) about some  $y_0^* \in \mathcal{H}_0^+$ . In the computed derivative below we do not use the star symbol, but  $y_0 = y_0^*$ ,  $y_1 = y_1^*(y_0^*)$ ,  $t_1 = t_1(y_0^*)$  and  $t_2 = t_2(y_1^*(y_0^*))$ . We get:

$$\begin{aligned} \frac{dy_2}{dy_0} = e^{-(t_1+t_2)} & \left( 1 - (y_0 + g_x^0 f_-^0) \frac{1 - e^{-t_1}}{h_{rd,x}^0 f_-^0 - e^{-t_1}(y_0 + g_x^0 f_-^0)} \right) \\ & \times \left( 1 - (y_1 + g_x^0 f_+^0) \frac{1 - e^{-t_2}}{h_{rd,x}^0 f_+^0 - e^{-t_2}(y_1 + g_x^0 f_+^0)} \right). \end{aligned} \quad (64)$$

For large enough times  $t_i = \mathcal{O}(1)$  ( $i = 1, 2$ ), we have

$$\begin{aligned} \frac{dy_2}{dy_0} \approx e^{-(t_1+t_2)} & \left( \frac{h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0}{h_{rd,x}^0 f_-^0} \right) \\ & \times \left( \frac{h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0}{h_{rd,x}^0 f_+^0} \right) \end{aligned} \quad (65)$$

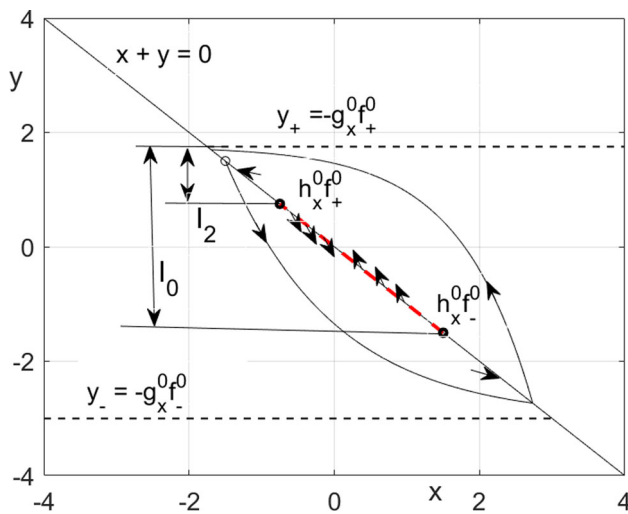
which is small and positive, and tends to 0 for  $t_i \rightarrow \infty$  ( $i = 1, 2$ ), by the conditions of Lemma 4.

We may also consider what happens for small times  $t_1$ , and  $t_2$ . Expanding the exponential terms in the leading order then gives:

$$\begin{aligned} \frac{dy_2}{dy_0} \approx (1 - (t_1 + t_2)) & \left( \frac{h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0}{h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0 + t_1(y_0 + g_x^0 f_-^0)} \right) \\ & \times \left( \frac{h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0}{h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0 + t_2(y_1 + g_x^0 f_+^0)} \right). \end{aligned} \quad (66)$$

Equivalently, we may express equation (66) as

$$\begin{aligned} \frac{dy_2}{dy_0} \approx (1 - (t_1 + t_2)) & \left( \frac{h_x^0 f_-^0 - y_0}{h_x^0 f_-^0 - y_0 + t_1(y_0 + g_x^0 f_-^0)} \right) \\ & \times \left( \frac{h_x^0 f_+^0 - y_1}{h_x^0 f_+^0 - y_1 + t_2(y_1 + g_x^0 f_+^0)} \right). \end{aligned}$$



**Fig. 2** Schematic depiction of phase space projected onto  $(x, y)$ -plane. The  $\dot{y} = 0$  nullclines, the switching manifold and repelling sliding subset are depicted. Compare with Case 1 in Fig. 1

We have that  $y_0 + g_x^0 f_-^0 > 0$ ,  $y_1 + g_x^0 f_+^0 < 0$ ,  $h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0 < 0$  and  $h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0 > 0$ . The necessary conditions for times  $t_1$  and  $t_2$  to be small is that the distances  $\|(x_1, y_1) - (x_0, y_0)\|$ , and  $\|(x_2, y_2) - (x_1, y_1)\|$  are small, which translates on the size of the sliding subset as measured by its euclidian length on the switching surface on the  $(x, y)$ -plane (since only the first component of the  $x$  state is of our interest, we ignore the indices). Moreover due to the uniqueness of the solutions generated by flows  $(X_\pm^t, Y_\pm^t)$ , we must have  $\frac{dy_2}{dy_0} > 0$ , and so it follows that either  $h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0 + t_1(y_0 + g_x^0 f_-^0) < 0 \wedge h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0 + t_2(y_1 + g_x^0 f_+^0) > 0$  or  $h_{rd,x}^0 f_-^0 - y_0 - g_x^0 f_-^0 + t_1(y_0 + g_x^0 f_-^0) > 0 \wedge h_{rd,x}^0 f_+^0 - y_1 - g_x^0 f_+^0 + t_2(y_1 + g_x^0 f_+^0) < 0$ . The right-hand side of (66) is continuous and bounded from the right for  $t_i \rightarrow 0$  ( $i = 1, 2$ ), and in the limit it is equal to 1. We will now summarise above findings and make them precise in the theorem below.

Without loss of generality, we set

$$h_{rd,x}^0 = [1, 0, \dots, 0], \quad g_y^0 = -1, \quad h_y^0 = 1 \quad (67)$$

with the first condition implying  $h_{rd,x}^0 x = x_0^1$ . We assume the initial condition  $(x_0^1, *, y_0) \in \partial\mathcal{H}_0$  where both vector fields  $F_+$  and  $F_-$  are transversal to  $\partial\mathcal{H}_0$  and point across  $\partial\mathcal{H}_0$  in the same direction. The “\*” symbol indicates the remaining  $n - 1$  states of the  $x$  state. In this setting, the Lie derivatives of  $h = h_{rd,x}^0 x + h_y^0 y$  in vector fields  $F_\pm$  at the initial point  $(x_0^1, *, y_0)$  have the same sign, which we choose to be negative, that is

$$\mathcal{L}_{F_\pm} h(x_0^1, *, y_0) = -y_0 + h_{rd,x}^0 f_\pm^0 - g_x^0 f_\pm^0 < 0, \quad (68)$$

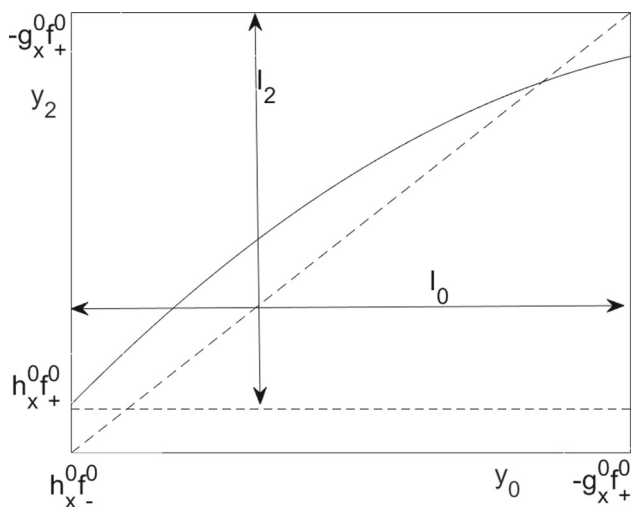
and the plus/minus symbol indicates states generated by vector fields  $F_\pm$ . By Lemma 4 there exists a return map  $y_0 \mapsto y_2$ , where  $y_2$  is now given by (63), and there exists solutions times  $t_i = t_i(y_0)$ , where  $i = 1, 2$ , which are positive solutions of equations (57) and (59).

**Theorem 7** Let  $x \in \mathbb{R}^n$  and  $y \in \mathbb{R}$ . Assume that conditions (67), (68) hold. Assume further that there exists a return map  $y_0 \mapsto y_2$  given by (63), and times  $t_i = t_i(y_0)$ , ( $i = 1, 2$ ) which are the smallest positive solutions of equations (57) and (59).

For  $y_0 \in (h_x^0 f_-^0, -g_x^0 f_+^0] = I_0$  the map (63) is monotonically increasing and differentiable from the right in  $y_0 \downarrow h_x^0 f_-^0$  and from the left when  $y_0 \uparrow -g_x^0 f_+^0$ , and hence an injection. It maps interval  $I_0$  within  $I_2 = (h_x^0 f_+^0, -g_x^0 f_-^0)$ . For sufficiently large  $t_i(y_0)$  ( $i = \pm 2$ ), the derivative  $dy_2/dy_0 \downarrow 0$ , and for sufficiently small  $t_i(y_0)$ ,  $dy_2/dy_0 \downarrow 1$ . It then follows that there exists at least one stable fixed point  $y^* \in I_2$  of map (63).

**Proof** Starting from any initial  $y_0 \in I_0$ , under the conditions of Lemma 4, the Implicit Function Theorem guarantees that there exist positive times  $t_1$  and  $t_2$  such that  $y_0$  is mapped onto  $y_2 \in I_2$ . Moreover the map (63) is continuously differentiable for any  $y_0 \in I_0$  in  $y_0$ ,  $t_1(y_0)$  and  $t_2(y_0)$  due to the continuity of (63) and (64) with respect to  $y_0$ ,  $t_1(y_0)$  and  $t_2(y_0)$ . Again, to see this we need to invoke the Implicit Function Theorem to determine the existence of continuous derivatives of  $t_1(y_0)$  and  $t_2(y_0)$  with respect to  $y_0$  for any  $y_0 \in I_0$ . The monotonicity of the map is ensured by the uniqueness of the flow solutions of vector fields  $F_\pm$  which make up the two trajectory segments which are concatenated together and map  $y_0$  into  $y_2$ . By the uniqueness of this concatenation any point starting at some  $y_0 \in I_0$  must be mapped to  $y_2$ , which lies to the left of  $y_0 \in I_2$ , and hence the monotonicity of the map is ensured. Moreover, the former ensures that  $dy_2/dy_0 > 0$  for any  $y \in I_0$ . Finally, we need to show the existence of at least one stable fixed point of map (63). We note that for any  $\varepsilon > 0$  a point  $h_x^0 f_-^0 + \varepsilon$  must be mapped above point  $h_x^0 f_+^0$  and hence the image of that point lies above the line  $y_2 = y_0$  (see Fig. 3). Since for sufficiently large  $t_i$  ( $i = 1, 2$ ), which corresponds to points close to  $-g_x^0 f_+^0$  within  $I_0$ , the slope  $dy_2/dy_0 \rightarrow 0$  by equation (65), the graph of map (63) must cross the line  $y_2 = y_0$  from  $y_2 > y_0$  towards  $y_2 < y_0$  at some point  $y^* \in I_0$ . This, in turn, implies that there exists a fixed point  $y^*$  in map (63) such that  $0 < dy_2/dy_0(y = y^*) < 1$ , and hence there exists at least one stable fixed point in (63). It now remains to show that for small  $t_i$  ( $i = 1, 2$ ),  $dy_2/dy_0 \rightarrow 1$ . Let  $y_0 = h_x^0 f_-^0 + \varepsilon_-$  and  $y_1 = h_x^0 f_+^0 - \varepsilon_+$  where  $\varepsilon_\pm$  are sufficiently small and positive. We expand  $h(x_1(t_1), y_1(t_1)) = 0$  and  $h(x_2(t_2), y_2(t_2)) = 0$  in  $t_1$  and  $t_2$ , and then solve

$$-y_0 + h_{rd,x}^0 f_-^0 t_1 + (1 - t_1 + \frac{1}{2} t_1^2)(y_0 + g_x^0 f_-^0) - g_x^0 f_-^0 = 0,$$



**Fig. 3** An example of a one-dimensional from  $I_0 \ni y_0 \mapsto y_2 \in I_2$

and

$$-y_1 + h_{rd,x}^0 f_+^0 t_2 + (1 - t_2 + \frac{1}{2} t_2^2)(y_1 + g_x^0 f_+^0) - g_x^0 f_+^0 = 0,$$

for  $t_1$  and  $t_2$  correspondingly in power series of  $\varepsilon_{\pm}$ . To leading order in  $\varepsilon_{\pm}$ , we find

$$t_1 = \frac{2\varepsilon_-}{h_{rd,x}^0 f_-^0} \quad (69)$$

$$t_2 = -\frac{2\varepsilon_+}{h_{rd,x}^0 f_+^0}. \quad (70)$$

Substituting (69) and (70) into (66) gives:

$$\begin{aligned} \frac{dy_2}{dy_0} &= \left(1 - \left(\frac{2\varepsilon_-}{h_{rd,x}^0 f_-^0} - \frac{2\varepsilon_+}{h_{rd,x}^0 f_+^0}\right)\right) \\ &\times \left(-\frac{1}{1 + (2\varepsilon_-)/h_{rd,x}^0 f_-^0}\right) \left(-\frac{1}{1 - (2\varepsilon_+)/h_{rd,x}^0 f_+^0}\right) \end{aligned} \quad (71)$$

Taking the limit  $\varepsilon_{\pm} \rightarrow 0$  gives:

$$\frac{dy_2}{dy_0} \uparrow 1 \quad \text{for } \varepsilon_{\pm} \downarrow 0. \quad (72)$$

□

We illustrate the above theorem in Fig. 2 and Fig. 3. In particular, in Fig. 2 we depict the phase space projected onto  $(x, y)$  components (where  $h_{rd,x} = x^1$  and we drop the upper index 1). In this setting,  $x + y = 0$  defines switching subspace  $\partial\mathcal{H}_0$ ,  $y_{\pm} = [g_y^{-1}]^0 g_x^0 f_{\pm}$  are the slow manifolds of vector fields  $F_{\pm}$  respectively. The red dashed line on  $\partial\mathcal{H}_0$  denotes the sliding subset within  $\partial\mathcal{H}_0$ , which is repelling. The boundaries of this set are also represented by  $h_{x}^0 f_{\pm}^0$ .

Compare with Scenario  $S_+ > T_+ > T_- > S_-$  in Fig. 1. An example of a trajectory starting at some  $y_0$  within  $I_0$  is shown to lead to some point  $y_2 \in I_2$  after a switch at  $S_0$ . A typical map  $I_0 \ni y_0 \mapsto y_2 \in I_2$  is then schematically depicted in Fig. 3. A stable fixed point corresponding to a period-1 cycle is clearly visible as the point of intersection between the curve representing the map with the line  $y = x$ .

## 7 Numerical examples

We will now present numerical examples which illustrate our findings. We expect that, typically, a sliding flow will be perturbed to a periodic behaviour with a drift.

**Example 1** We set  $h_{rd,x}^0 = [1 \ 0]$ ,  $f_-^0 = [1.5 \ 1]^T$ ,  $f_+^0 = [-1 \ -0.5]^T$ ,  $[g_y^0]^{-1} g_x^0 = [-1 \ -1.5]$ ,  $-g_y^0 = 2$ ,  $h_y^0 = 1$ , and  $h_x^0 = h_{rd,x}^0 + h_y^0 [g_y^0]^{-1} g_x^0 = [0, -1.5]$ . The initial conditions are  $(x_1(0), x_2(0), y(0)) = (-0.25, 0, 0.5)$  (full system),  $(x_1(0), x_2(0)) = (0, 0)$  (reduced system). The above set of matrix values correspond to the following leading order, reduced:

$$\dot{x} = \begin{cases} \begin{pmatrix} -1 \\ -0.5 \end{pmatrix} & \text{if } x_1 > 0, \\ \begin{pmatrix} 1.5 \\ 1 \end{pmatrix} & \text{if } x_1 < 0, \end{cases} \quad (73)$$

$$0 = y, \quad (74)$$

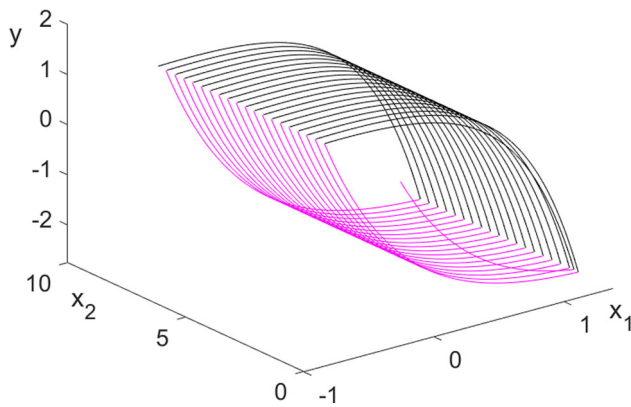
and full system:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{cases} \begin{bmatrix} \begin{pmatrix} -1 \\ -0.5 \end{pmatrix} \\ -2y + 4.5 \end{bmatrix} & \text{if } x_1 + 0.5y > 0, \\ \begin{bmatrix} \begin{pmatrix} 1.5 \\ 1 \end{pmatrix} \\ -2y - 6 \end{bmatrix} & \text{if } x_1 + 0.5y < 0, \end{cases} \quad (75)$$

We verify that  $S_+ = h_y^0 [g_y^0]^{-1} g_x^0 f_+^0 = 1.75 > T_+ = h_x^0 f_+^0 = 0.75 > T_- = h_x^0 f_-^0 = -1.5 > S_- = h_y^0 [g_y^0]^{-1} g_x^0 f_-^0 = -3$ . We thus confirm that the case of repelling sliding must take place in the full system.

A representative trajectory in phase space of the 3-dimensional slow-fast system is depicted in Fig. 4. We find that  $t_1$  and  $t_2$  converge to values  $t_1 = 1.4857$  and  $t_2 = 2.2285$  and hence define a stable period-one orbit for which we find:

$$\alpha = -\frac{h_{rd,x}^0 f_+^0}{h_{rd,x}^0 (f_-^0 - f_+^0)}$$



**Fig. 4** Representative trajectory in phase space of the full system (32) for numerical values set to:  $h_{\text{rd},x}^0 = [1 \ 0]$ ,  $f_+^0 = [-1 \ -1.5]^T$ ,  $f_-^0 = [1.5 \ 1]^T$ ,  $[g_y^0]^{-1} g_x^0 = [-1 \ -1.5]$ ,  $-g_y^0 = 2$ ,  $h_y^0 = 1$ , and the initial conditions  $(x_1(0), x_2(0), y(0)) = (-0.25, 0, 0.5)$  (full system),  $(x_1(0), x_2(0)) = (0, 0)$  (reduced system - not depicted)

$$= \frac{t_1}{t_1 + t_2} = 0.4.$$

**Example 2** As a second example we consider the modified dry-friction oscillator model introduced in [18]. It is given by

$$\dot{x}_1 = x_2, \quad (76)$$

$$\dot{x}_2 = -\frac{K}{M}x_1 + \frac{1}{M}f_0 \text{sgn}h(x, w), \quad (77)$$

$$\varepsilon \dot{y} = -y + v_0 + \gamma x_2 - \varepsilon \gamma \frac{K}{M}x_1 \quad (78)$$

(with positive  $K, M, f_0, c_0, \varepsilon$  and  $\gamma < 0$ ), where  $h(x, y) = -(1 + \gamma)x_2 + y$ . Thus, the critical manifold is given by  $y = v_0 + \gamma x_2$ . The reduced system ( $\varepsilon = 0$  in (78)) is then

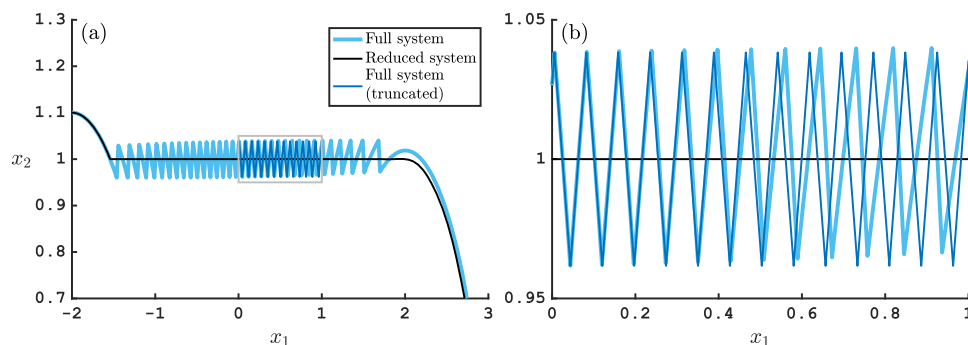
given by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{K}{M}x_1 + \frac{1}{M}f_0 \text{sgn}(v_0 - x_2), \end{aligned} \quad (79)$$

In the reduced system (79), attracting sliding takes place along the subset  $\{(x_1, x_2) : x_2 = v_0, |x_1| < f_0/K\}$  of the switching set  $\{(x_1, x_2) : x_2 = v_0\}$ . Let  $K = M = 1$ ,  $f_0 = 2$ ,  $\gamma = -2$ ,  $v_0 = 1$ ,  $\varepsilon = 0.01$ , and consider dynamics near the point  $(x_{1,0}, x_{2,0}, y_0) = (0, 1, -1)$  on the switching surface of the reduced system (79), intersected with the critical manifold. In this point and for these parameters we have  $h_x^0 = (0, 1)$ ,  $h_y^0 = 1$ ,  $(g_y^0)^{-1} = -1$ ,  $g_x^0 = (0, -2)$ ,  $f_{\pm} = (1, \pm 2)^T$ , such that

$$S_- = -4 < T_- = -1 < T_+ = 1 < S_+ = 4.$$

Thus, we expect that the full system (76)–(78) will exhibit rapid switching near the sliding trajectory of the reduced system. This is depicted in the phase space projection in Fig. 5. The trajectory of the reduced system starting from  $(-2, 1.1)$  is the black curve. It enters sliding near  $x_1 = -1.5$  and slides for increasing  $x_1$  up to  $x_1 = 2$ , passing through the point  $(0, 1)$  we consider. The full system follows the reduced system closely for the initial parts of the trajectory (where the flow  $f_+$  acts), as guaranteed by Fenichel's theory [5] for our small  $\varepsilon$ . Instead of sliding, the full system shows small oscillations around the projected switching surface. The trajectory of the truncated full system after rescaling back by  $(x, y) \mapsto (x_0, y_0) + (x, y)/\varepsilon$  (dark blue curve) approximates the full dynamics accurately near the expansion point  $(x_0, y_0) = (0, 1, -1)$ , as the zoom in Figure 5(b) shows.



**Fig. 5** Sliding and switching trajectories for modified dry-friction oscillator (76)–(78) projected into  $(x_1, x_2)$  plane. The black curve is the trajectory of the reduced system (79) from initial condition  $(-2, 1.1)$ . It has a sliding segment along  $x_2 = v_0$ . The light blue curve is the trajectory of the full system, starting from  $(-2, 1.1, -1.2)$ , showing oscillations around the sliding surface. The dark blue curve is the trajectory of the full system's truncated expansion at center point

$(x_0, y_0) = (0, 1, -1)$ , starting at  $(0, 1.027, -1.027) = (x_0, y_c(x_0)) + \varepsilon(0, 2.7, -1.7)$ , after scaling back by  $(x, y) \mapsto (x_0, y_0) + \varepsilon(x, y)$ . Panel (a): Projection onto  $(x_1, x_2)$  co-ordinate plane. Panel (b): zoom-in to neighborhood of discontinuity boundary  $x_2 = v_0$  of reduced system. System parameters:  $K = M = 1$ ,  $f_0 = 2$ ,  $\gamma = -2$ ,  $v_0 = 1$ ,  $\varepsilon = 0.01$ . Computed with solver *filippov* for Matlab [17]

The oscillations of the truncated and the full system match closely close to  $x_1 = 0$ . As the trajectory moves on along the discontinuity (for larger  $x_1$ ) the accuracy of the truncated system diminishes, because the trajectory is now far away from the expansion point (contrast the difference in shape and period of oscillations in dark and bright blue for  $x_1 \in [0, 0.3]$  with those for  $x_1 > 0.5$ ). The trajectory of the truncated full system is periodic in  $x_2$  (and  $y$ , not shown), while the properties of the oscillations drift for the full-system trajectory as  $x_1$  changes. Note that  $x_1$  is proportional to time since for sliding motion  $\dot{x}_1 = x_2 = v_0 = 1$ , such that the time moves forward along the trajectory from left to right, where the oscillations of the full system proceed with the same speed as the sliding motion of the reduced system. Numerics of the truncated and full systems is conducted using the solver `filippov` for Matlab [17].

## 8 Conclusions

In the current work we investigate the effects of singular perturbation on the sliding dynamics present in a Filippov system. In particular, we consider the case when the sliding flow corresponds to the dynamics of the reduced Filippov system. We then consider what is the effect of stable singular perturbations on that flow. Generically, for  $n$ -dimensional slow dynamics and  $m$ -dimensional fast dynamics, due to singular perturbations we distinguish five qualitatively distinct scenarios as we classify them in Sect. 4 (see Table 1 and Fig. 1) by means of a sequence of trajectory segments generated by the perturbed flow. This is our first main result. Namely, due to singular perturbations of the sliding reduced flow we may observe: a) switchings between vector fields  $F_{\pm}$  along the sliding flow of the reduced system (Scenario  $S_- < T_- < T_+ < S_+$ ); b) sliding or sliding segments possibly interspersed with switchings to  $F_{\pm}$  (Scenario  $S_- < T_+ < T_- < S_+$ ); c) sliding or sliding segments possibly interspersed with switchings to  $F_-$  (Scenario  $T_+ < S_- < T_- < S_+$ ); d) sliding or sliding segments possibly interspersed with switchings to  $F_+$  (Scenario  $S_- < T_+ < S_+ < T_-$ ); e) sliding (Scenarios:  $T_+ < S_- < S_+ < T_-$ ,  $T_+ < S_+ < S_- < T_-$ ). In Scenarios b)-d) the exact dynamics will depend on the stability properties of the sliding flow in the full system. We may further classify scenarios a) to e) into two cases. Namely, case one which corresponds to scenario a) which is characterised by the presence of the repelling sliding set in the perturbed systems, and case two which corresponds to scenarios b) to e) which is characterised by the presence of an attracting sliding set in the perturbed system.

In the current paper case one is then investigated in detail in Sects. 5 and 6. The main results of Sect. 5, encapsulated in the summary as Theorem 3, states that stable singular per-

turbations of the sliding flow with the fast dynamics having dimension  $m > 1$  results in perturbation of  $\mathcal{O}(\varepsilon)$  that depends on  $t$  and is only continuous, so it can be rapidly fluctuating. This, in turn, may imply the existence of micro chaotic dynamics as reported in different contexts in Filippov type systems [4, 9, 10]. The special case of  $m = 1$  is then treated in Sect. 6. In such case, stable singular perturbation implies regular perturbation of  $\mathcal{O}(\varepsilon)$  of the full flow, and hence no complex dynamics, even on a micro scale, is possible. We illustrate our findings by two numerical examples in Sect. 7. The future work is focused on investigating case two and applying our analytical findings to models of relevance to applications.

To summarise: case one may be considered as “safe” in the sense that the singular perturbation may not significantly alter system dynamics. On the other hand, case two may turn out to be “unsafe” in the sense that, for example, a significant change in the direction of the perturbed flow cannot be excluded. Of course, it remains to be investigated if there is such a possibility.

**Author Contributions** Both authors contributed equally to theoretical analysis, performance of illustrative computations and write-up of manuscript.

**Funding** The authors have not disclosed any funding.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no Conflict of interest.

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